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1 Summary in Romanian

1.1 Descrierea temei

Datorită evoluției rapide a tehnologiei, camerele cu termoviziune au devenit o unelată esențială în mai multe industrii, oferind soluții eficiente și inovatoare pentru analiza și detectarea distribuției căldurii. Ele captează radiațiile infraroșii emise de obicete, urmând mai apoi să transforme această informație în imagini vizuale. Prin acest proces camerele cu termoviziune permit utilizatorului să observe diferențe și anomalii de temperatură într-un mod detaliat și precis. Aceste dispozitive sunt prezente în mai multe domenii precum cel industrial, cel de securitate, cel de agricultură și cel medical, îmbunătățind eficiența operațională, siguranța și economisind resusrse. Datorită adaptabilității acestor dispozitive și a mediul larg de aplicații în care pot fi folosite, detectarea variaților de temperatură utilizând camere cu termoviziune a devenit esențială într-o lume în care eficiența și precizia sunt esențiale.

Prin intermediul termografiei, analiza se realizează fară ca instrumentul de măsură să aibe contact direct cu obiectul testat. Experimentele se fac într-o maniera non-distructivă cu ajutorul razelor infraroșii care acționează pe spectrul electromagnetic la lungimiile de bandă de 2-5 µm și 8-12 µm. Aceste raze nu sunt vizibile cu ochiul liber, ceea ce ajută utilizatorul în detectarea modificărilor de temperatură. Termoviziunea permite astfel detectarea modificărilor de temperatură a corpurilor calde cât și a celor reci.

1.2 Descrierea concisă a sistemului

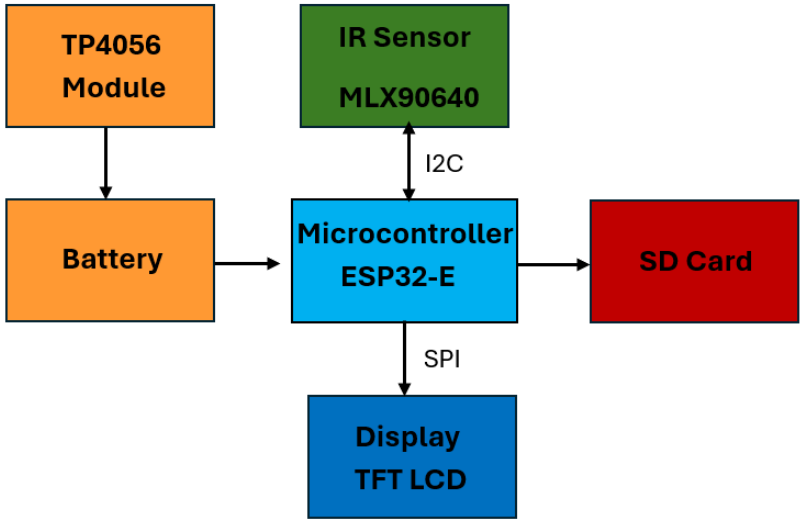


Figura 1. Schema bloc

În Figura 1, este prezentată schema block a sistemului format din microcontrollerul ESP32-E, care procesează datele primite de la modulul utilizat pentru citirea temperaturilor, după care, valorile citite, urmează să fie afișate pe ecranul unui display LCD, pe o hartă termică. Acest proces se întamplă în timp real iar imaginile pot fi salvate pe un card SD atașat sistemului. Întregul sistem este alimentat de

o baterie reîncărcabilă care oferă longevitate dispozitivului. Pentru partea de software, s-a realizat un program dedicat special pentru îndeplinirea obiectivelor prevăzute în acest proiect.

1.3 Proiectarea sistemului

Din punct de vedere al componentelor hardware folosite în acest sistem microcontrollerul FireBeetle 2 ESP32-E a fost folosit, deoarce ultilizează cea mai recentă variantă de microchip de pe piață, de la Espressif. Acesta oferă 16MB de stocare flash, 2MB de PSRAM și poate funcționa la frecvențe de până la 240MHz. Acest microcontroller cuprinde și o antenă care oferă conexiune prin Bluetooth și Wi-Fi. De asemenea pentru o ușoară comincare cu restul componentelor acest microcontroller este prevăzut cu 11 pini analogici, 17 pini digitali, 3 interfețe UART, 1 interfață SPI, 1 interfață I2C, 1 interfață I2S și 2 interfețe DAC.

Ca și tensiune de alimentare, acesta a fost conectat la o baterie reîncărcabilă de 500mAh prin portul PH2.0 de pe placă. Bateria poate fi reîncărcată cu un cablu de tip micro-USB. Pentru a face acest lucru posibil, a fost folosit modulul TP4056 care este prevăzut și cu protecție pentru supra-încărcarea bateriei. Acest modul ofera utilizatorului informații despre starea bateriei prin intremediul a două LED-uri: unul pentru momentul în care bateria este în process de încărcare (roșu) și unul pentru când procesul de încărcare este terminat iar bateria este complet alimentă (albastru).

Senzorul de temperature folosit este MLX90640 de la Adafruit. El este setat să citească temperaturile într-o formă de matrice asemănătoare unei table de șah. Cu o rezoluție de 32x24 pixeli acesta formează o hartă termală (Figura 2) prin fiecare pixel care înregistrează câte o măsurătoare, totalul fiind de 768 de măsurători care se repată constant cu frecvența de 2 Hz. Această frecvență a fost aleasă pentru a lua măsurători cât mai precise. La viteza de 2 imagini pe secundă și cu procesarea pe 18 biți a informației de către convertorul analog-digital (ADC) s-au obținut cele mai bune rezultate din punct de vedere al acurateței în măsurători. Acesta comunică cu microcontrollerul utilizând protocolul I2C pe 2 fire (SDA și SCL) și oferă citiri de temperatură în intervalul -40°C până la 300°C. Cu un câmp de vedere mare de 110°x75° și o toleranță la zgomot extern ridicată, acest modul se remarcă ca fiind printre cele mai profesionale și bune opțiuni pentru acest sistem.

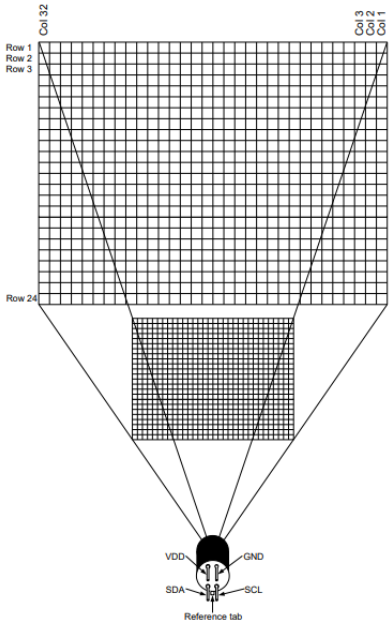


Figura 2. Matricea prin care senzorul realizează citirile de temperatură

Pentru afișarea în timp real a măsurătorilor realizate de către senzorul prezentat anterior un display TFT LCD a fost ales. Acesta are o dimensiune de puțin peste 7 centimetrii pe diagonală. Prin intermediul technologiei „thin film transistor liquid crystal display”, acesta afișează pe ecran măsurătorile într-un mod colorat pentru o ușoară interpreare a rezultatelor. Rezoluția ecranului este de 240x320 pixeli. Prin intermediul microcontrollerului intergat în display, ILI9341, display-ul furnizat de Adafruit Industries oferă două posibilități de conectare: cea pe 8 biți și SPI. Protocolul de comunicare folosit în acest sistem este cel SPI deoarece, prin acesta s-a realizat și conexiunea cu portul de card SD. Cu apăsarea unui buton imaginile de pe ecran pot fi trimise spre cardul SD pentru stocare și pentru a fi ulterior analizate mai în detaliu. Pentru stocarea imaginiilor pe cardul de memorie extern un buton a fost plasat în stânga ecranului pentru a permite uilizatorului să capteze exact imaginile pe care și le dorește.

Pentru a reduce spațiul ocupat de componente și pentru a oferii un aspect mai curat al sistemului un PCB a fost făcut special pentru acest dispozitiv. Acesta realizează conexiunile între toate dispozitivele și asigură comunicarea între microcontroller și perifericele acestuia. În special pentru protocolul de comunicare SPI traseele aferente au fost separate de restul traseelor pentru a asigura integritatea semnalului. Routarea a fost realizată pe 2 straturi (top si bottom) pentru a ecnonomsii cât mai mult spațiu. De asemenea, PCB ul este prevăzut cu 4 găuri de montare (câte una în fiecare colț) pentru a fi ușor de integrat într-o carcasă 3D facută special pentru acest sistem.

În interiorul carcasei printate 3D pe partea de bottom sunt plasate componentele la care utilizatorul nu are acces direct precum: microcontrollerul, senzorul de temperatură, bateria și modulul de încărcare, carcasa având o deschidere doar pentru senzor. Astfel pe partea de top, singurele componente prezente sunt cele cu care interacționează direct utilizatorul: ecranul și butonul. În partea de sus a carcasei o deschidere pentru portul de încărcare și releul pentru a porni sau opri dispozitivul sunt prezente, iar în partea de jos a fost prevăzută o deshidere protejată pentru accesul la cardul SD. În Figura 3 se poate observa structura internă a dispozitivului.

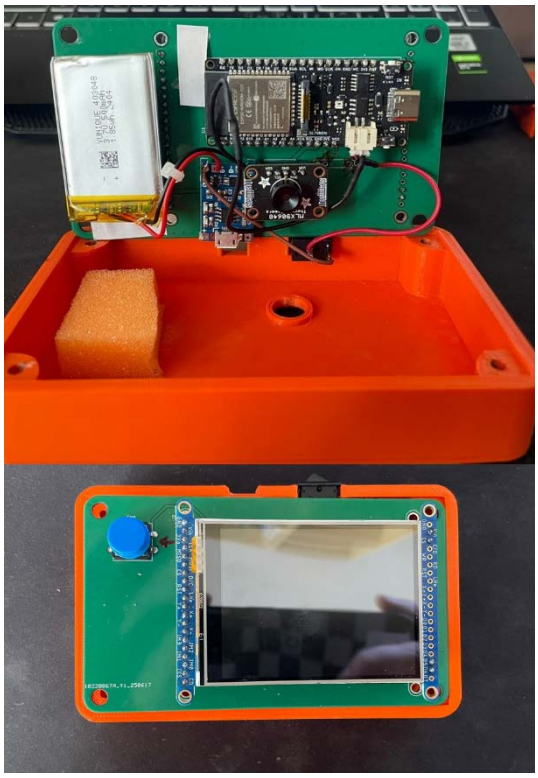


Figura 3. Interiorul carcasei

În partea de sus din Figura 3, se observă componentele descrise pe partea de „bottom” cu deschiderea pentru senzor, iar în partea de jos componentele de pe „top” la care are acces direct utilizatorul.

Pentru implementarea software a sistemului s-a utilizat aplicația Arduino IDE compatibilă cu microcontrollerul ales. În acest proces calculul de temperatură a fost realizat cu ajutorul librariilor integrate în aplicație pentru a obține cele mai bune rezultate cu componentele folosite. Astfel, procedee precum setarea gradientelor de culoare au fost implementate în cod, precum și modul de funcționare al senzorului de temperatură. Mai multe procedee de ajustare sunt prezente în aceasta parte pentru a orienta și așeza imaginile in direcția potrivită. Pentru salvarea hărților termale pe cardul de memorie externă o mărire a imaginiilor a fost aplicată. Ordinea proceselor este prezentată și în Figura 4 pentru o înțelegere mai bună.

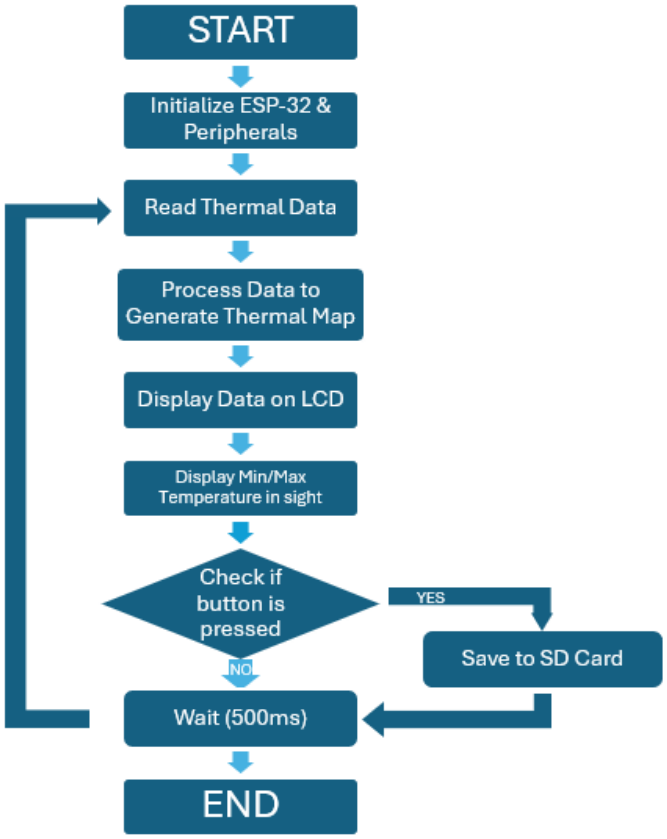


Figura 4. Organigrama software a sistemului

După pornirea dispozitivului se inițializează microcontrollerul ESP32 și perifericele. Senzorul de temperatură realizează măsurătorile după care, se realizează harta termală afișată pe ecran. De asemenea pe ecran sunt prezente 2 ferestre pentru afișarea temperaturii maxime si minime din harta termală. După care dacă butonul de salvare al imaginii este apăsat imaginea se salvează. În caz contrar procesul se întoarce la starea de citire pentru a realiza o măsurătoare nouă. Tot în partea software, codul utilizat prevede o întârziere de 500ms pentru a nu supra-încărca un proces. Astfel citiriile se întâmplă constant pentru a putea urmării modificările de temperatură în timp real până când dispozitivul este oprit prin acționarea releului.

1.4 Testarea sistemului

Pentru testarea dispozitivului, acesta a fost pus în mai multe medii și pe subiecte diferite. De asemenea, rezultatele au fost comparate cu cele obținute cu ajutorul unei unelete profesionale utilizată pentru măsurarea temperaturilor, prin infraroșu. Dispozitivul folosit pentru verificarea celui prezentat este camera cu termoviziune THT200 produsă de HT Instruments. Cu rezoluție și acuratețe mult mai mari dar și un preț de aproximativ 10 ori mai scump decât dispozitivul implementat, rezultatele demonstrează funcționalitatea bună a sistemului implementat.

Pentru obiecte mici precum microchipuri eroarea senzorului MLX90640 de ±2°C este mai des întalnită decat în alte situații. Chiar și așa rezultatul nu este departe de cel înregistrat de camera THT200. Pentru 2 componente electrice apropiate una de cealaltă, din cauza campului de viziune a senzorului MLX90640 este nevoie ca acesta să se apropie mai mult de subiectul studiat, pentru a obține rezultate cât mai precise. În cazul obiectelor de dimensiuni mai mari testele demonstrează că sistemul implementat oferă rezultate bune. În testul de detecție al unui obiect rece senzorul folosit citește o valoare exactă pe când camera uneltei THT200 citește și afișează o valoare de sub 0°C. De asemenea, pentru testarea prezenței unei persoane într-o încăpere sistemul realizat oferă un rezultat mult mai bun decât camera profesională. Astfel, dispozitivul realizat demonstrează, acuratețe, precizie dar și versatilitate.

1.5 Concluzii

Sistemul este ușor de utlizat, cu un cost mai mic decat produsele deja existente și demonstrează versatilitate, precizie și acuratețe. Întregul sistem poate fi îmbunătățit cu ușurință pentru a atinge o aplicabilitate și mai mare. Astfel de îmbunătățiri viitoare pot fi realizate atât pe partea de hardware prin înlocuirea componentelor cu unele mai performante, dar și pe partea software, prin îmbunătățirea cursivității codului și adăugarea de întreruperi software. Prin utilizarea ESP32 sistemul poate fi adăugat în IoT (Internet of Things) pentru a fi utilizat la scară largă. De asemenea, se pot adăuga conexiuni wireless pentru acces de la distanță a dispozitivului printr-o interfață web.

2 Work planning

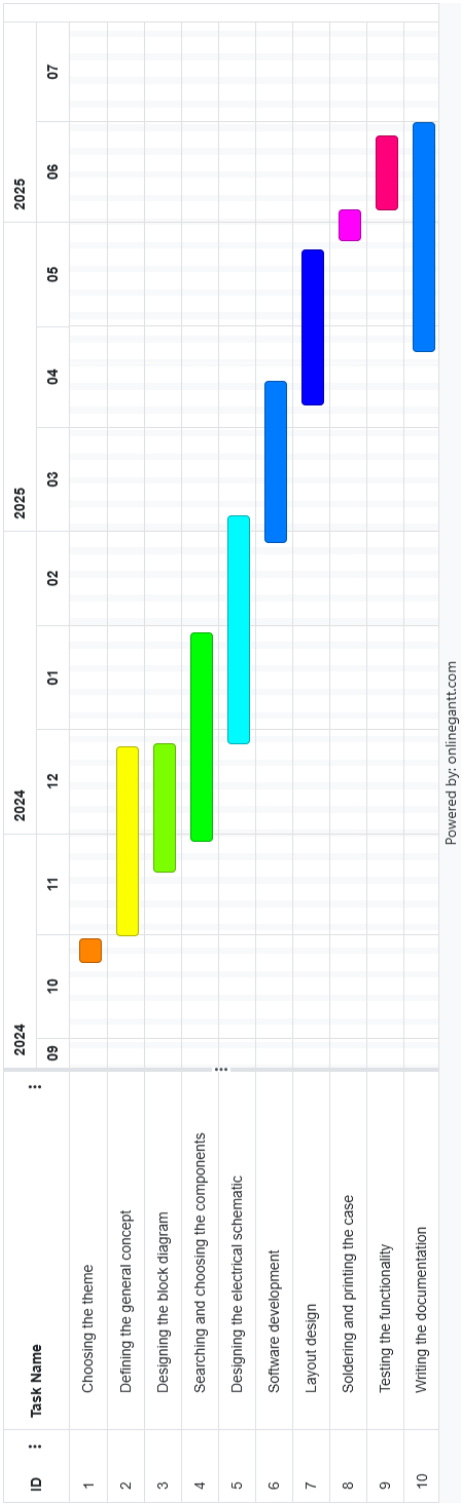


Figura 5. Gantt diagram of the work planning

3 State of the art

3.1 Introduction

Thermal imaging cameras can provide the user with high-quality thermal images for industrial applications. They can be used to discover faults, failures and flaws before they become a problem. By noticing temperature changes with the use of infrared technologies, thermal imaging aids in finding liquid or gas leaks, electrical inspections, troubleshooting and maintenance.

The thermal camera is a non-contact measurement tool that captures infrared energy naturally emitted by any object. They convert the infrared radiation into a two-dimensional temperature-coded image that is displayed to the user as a thermal map revealing heat signatures of the analyzed subject. These types of cameras can be used for a broad range of applications such as remote inspection, tests on objects that are obscured due to poor visibility, surveillance and security.

The purpose of this thesis is to design a cost efficient portable thermal imaging system with a compact design and real-time display accessible to both professional and hobbyists' fields.

3.2 Thermography

Fault localization and identification is essential in failure analysis, especially when testing a product. General methods for fault localization include infra-red optical beam induced resistance charge (IR-OBRICH), photon emission microscopy (PEM), lock-in thermography (LIT) and voltage contrast to perform a physical failure analysis precisely. Nevertheless, these methods are mostly performed on components and parts that may require delayering or decapsulation while for passive devices, printed circuit boards (PCBs) and advanced packaging chips thermography may be a better solution for failure localization [1].

In most cases, failures can be categorized as: high resistance, leakage, open, short and functional failure. The heat that is generated as soon as a voltage is applied and current flows through the resistive elements of a device, those elements heat up with a proportion following the formula:

$$P = I^2 \times R \tag{1}$$

"P" being the power or the rate of heat generation, "I" the current flowing through the resistive elements and R the resistance. This heat can be seen as thermal radiation, thermal conduction or themal convection. For the heat to be displayed two bands are used most of the time for remote sensing applications. These bands wavelengths range from 3 µm to 5 µm and from 8 µm to 12 µm for a higher transmission flow in the air (the wavelengths between 5 µm to 8 µm the infrared radiation suffers very strong absorption from water vapor and carbon dioxides). These wavelengths are displayed in Figure 1 [1][2].

The basic thermographic cameras are composed of a lens that is transparent to infrared radiation and a matrix M x N made out of sensitive elements that transform this energy into an electrical signal. If a object signals a defect it will cause a change of color in the matrix and in the thermal pattern giving the advantage of not needing an exact measurement but rather the result is given by a comparative analysis. As a result, detecting the temperature maps are a useful tool to point out the power distribution of an electronic device.

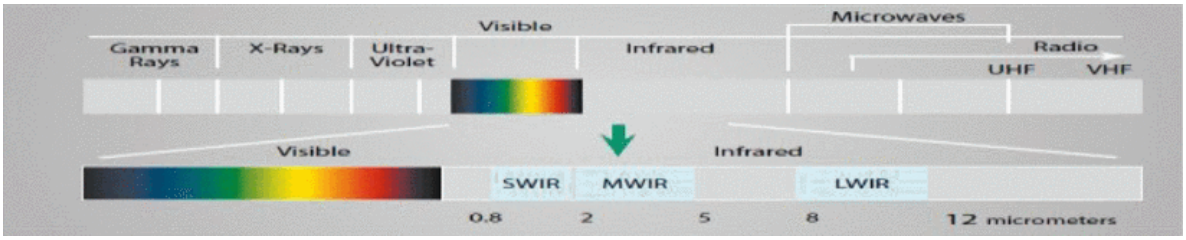


Figure 1. Electromagnetic spectrum showing the infrared band and highlighting the short wave IR wavelenght of 3 to 5 µm and Long Wave IR from 8 to 12 µm [2]

One of the most limiting factors, in the present times, in the performance and degradation of electronics such as power modules is the thermal dissipation, therefore, active condition monitoring is a great way to improve the electronics lifetime. Nowadays, the industry has gradually shifted from oversizing components like conductors, semiconductors and heat sinks in order to handle more power and ensure safety and reliability without having to active monitor or control the active state of a device. By not using this old method, especially in power electronics, a smart power actuator and senor interface were introduced that can manage the thermal characteristics in real time. By actively monitoring the thermal changes of a system it is possible to schedule maintenance before serios faults may occur and allows increased safety margins, therefore enhancing curative actions [3].

3.3 Thermography around us

As technology evolves around us, more and more types of sensors made their ways into everyone's lives. Among them, the infrared thermal imaging sensors grew in popularity due to their wide range of applications. For example, they have been present in the military field for ballistic missile defense, in civilian field, for monitoring transformers, switchgears and transmission lines, in the power industry, in the security field because the infrared sensor can detect the presence of a person even in low visibility conditions such as low light exposure, intentional fog, steam or dust generation and physical tampering with the sensor lens. For the medical field, the range of applications is even wider, thermal imaging being used for detecting skin diseases, tumors, vascular diseases and changes in the human body temperature. The request of thermal imaging devices has also risen rapidly after the COVID-19 outbreak [4]. No matter where this type of camera or sensor is used, the infrared thermography is a contactless, non-destructive method used for quantitative thermal measurements [5]. Thanks to the increased accessibility of the infrared sensors and the increasing performance of Readout Integrated Circuit (ROIC), thermal sensing cameras have completely replaced the single sensor by using a fast transient infrared thermal analysis with MOSFETS [6].

3.3.1 Thermography in odometry

A recent study in odometry shows how using directly thermal RGB frames in order to aid in accuracy of inertial measurement units (IMUs), accelerometers, gyroscopes and LiDARs. Because IMUs are known to suffer from growing estimation errors due to slowly varying nonidealities and biases they are often combined with visual data (e.g. an RGB camera), therefore creating a visual-inertial odometry system (VIO). By doing so, the VIO was able to successfully be a part of many navigational systems providing more precise positioning, localization and mapping setups, generating new maps. But by using just the RGB camera in the visual-inertial odometry system an issue came

when looking at the lighting conditions. As a solution, radars, high-dynamic-range cameras (HDR), and LiDARs were used as sensing modalities in low-light conditions. Between these solutions, a thermal camera usage became more popular because of its power consumption, size and robustness and also because it was a very good competitor to the bulkiness of the LiDAR, the price of the HDR and the power consumption of the radars. Therefore, the thermal sensing camera is both power-efficient and size efficient also offering a high resolution in many applications. [7]

3.3.2 Thermography in drone surveillance

Another study discusses the usage of thermal imaging cameras in surveillance and security. The paper presents the process of continuous observation of an area or a person by using a custom-designed drone which incorporates a dual-spectrum camera that can provide data even in dark conditions and areas where fog and rain condensation is present without other external illumination. The system was able to identify the presence of a military person using the infrared technology attached to it and to differentiate allies and intruders by implementing an RFID sensor. Because all objects warmer than -273K (absolute zero temperature) emit IR radiation, this can be detected by the heat sensing cameras. The heat-scan can then be converted into frames with different shades of gray which represent variations in temperatures. Especially at night-time when animals, humans and vehicles are warmer than the ambient temperature the IR camera can still provide valuable data. The images produced by this system are called thermograms and the detection of a specific object is done by using different algorithms in order to display the wanted subject. Therefore, additional security and safety can be provided by using this system that incorporates the thermal sensing device [8].

3.3.3 Thermography in Oil Spill detection on Sea Surface

As we take a look at the marine ecological environment, we can see how it can affect human healthcare. Especially hazards and accidents are a serious threat nowadays when observing the amount of marine oil-spill accidents. These types of incidents can be detected and identified by using remote sensing. A study shows how thermal infrared imaging remote sensing images of the incidents were obtained by using a unmanned aerial vehicle (UAV) with a thermal imaging camera. This is a helpful method in accurate detection of oil spills in the marine transportation and oil extraction industries. By doing so, effective and rapid treatments of the sea surface can be started. This method is a faster one than the older ones because of the UAV remote sensing technology in the marine oil spills domain [9].

The current problems that may occur when doing a analysis on the sea surface are the subjective and environmental factors in the identification of the oil film. The oil film’s thickness absolute value is directly proportional to the level of brightness [9].

After implementing the thermal IR sensing detection on a experimental set-up, the image obtained shows how the temperature increases with the thickness of the oil in each container used to simulate various types of oil in real marine scenarios. Therefore, once again the thermal imaging camera proves to be useful in another environment.

3.3.4 Working principle

This technology detects the temperature distribution across an area of a device by using a custom radiometric 2D measurement system that enables the user to see heat maps of the device surface. By capturing the two-dimensional thermal maps, we can observe how an electronic device is energized

and how the heat rises locally, and it mirrors the principal power and current densities, therefore the infrared imaging reveals how heat is distributed across the device.

Another technique commonly used for thermal inspection in a contactless manner for the heat distribution on semiconductors and integrated circuits is the thermo-reflectance imaging. By using this method, we can achieve a very high accuracy when looking at the temporal and spatial resolution. This procedure requires a longer and accurate calibration process because it is based on the reflectivity coefficient, and it is strongly dependent on the experimental conditions and the wavelength of a laser source. Additionally, this coefficient needs to be large enough to ensure a suitable signal to noise ratio [5].

To take a look into the internal structure of the thermal imaging sensor most of them make use of a bandgap voltage and a proportional to absolute temperature (PTAT) sensor. This sensor has the output voltage directly proportional to the absolute temperature, and it is commonly present in integrated circuits because of their high sensitivity and their capability to accurately measure temperature variations. The array of M x N sensitive elements are the actual infrared detecting elements, each pixel's resistance changing with the incident infrared power. As for the analog front end, each pixel is amplified and the analog to digital converters (ADCs), convert each pixel's voltage into digital data, that is then stored into the internal random-access memory (RAM). The EEPROM of the sensor holds the data about the factory calibration of the device. Of course, most sensors use standard power rails for the whole module to work properly and a communication protocol to ensure data transfer between the sensor module and the processing unit [4].

The images provided by such a system can always be enhanced in two ways: frequency domain and spatial methods. The most popular approach to image enhancement is histogram equalization. This method extends the dynamic range of the image in separate regions, causing alterations in the overall tone of the image. As for the frequency domain enhancement, the image undergoes modifications in magnitudes, altering the frequency fulfilment of the image. These modifications use frequency transforms including Fourier transforms and discrete cosine transforms (DCT). By using any of these two methods, the result comes to be successful in getting more contrast and detailed thermal images [10].

4. Theoretical Fundamentals

For the theoretical background necessary to understand the technology used in the thermal sensing camera we must take a closer look at the system characteristics. In recent years, we can observe a lot more interest in implementing smart sensing systems in various industries such as automotive, naval, agriculture, safety, aerospace etc. This project focuses on developing a thermal sensing camera project, powered by a battery with real-time display. The system uses a special sensor that measures the quantity of infrared radiation of surrounding objects. All the components are integrated upon a Printed Circuit Board (PCB) and a custom 3D printed encloser. In this chapter we will analyze every component characteristics.

4.1 Power management

In such projects where portability of the device is key, of course, power management is a important part. Nowadays, the market has a variety of options when it comes to portable power delivery. It is so that we don't have to plug into an outlet any device and especially for smaller voltages there is no need to add step-down converters that can get expensive in the manufacturing process. Some of the most popular options are the rechargeable batteries with the most popular ones being the Lithium Polymer and the Lithium-ion variants. They come with a compact design, high energy density, lightweight and the property of being rechargeable. Also, they are easy to integrate with modules that limit the charging capability of the battery and ensure safety when they are used.

4.1.1 LiPo batteries

In order to get the best combination of size, reliability, performance and weight, the Lithium Polymer batteries provide the best quality when it comes to applications such as drones, robotics, thermal sensing cameras and other devices. Standing at a lighter weight than other batteries and with the advantage of being in different shapes than the traditional cylinder batteries and the flat shapes are making them easier to be placed in the case of a product. With the high energy density, they can store a large amount of energy into a small volume. This can be also seen in our smartphones, tablets or anywhere where the weight is limited but the power demand is high. With a nominal voltage of 3.7V but when connected in series they can reach higher voltages. Due to their high discharge rates, they are a good option in applications with high-performance tools [11].

As for the safety aspects of the LiPo batteries, it is known that they are prone to damage and therefore require the user to be careful with them when storing and using them, because most of the failures happen due to external factors. Paired with the right electronic module the problems with overcharging can be reduced significantly.

4.2 The Microcontroller

For the development of embedded systems, the choice of a capable, performant and functional microcontroller is one of the hardest since the market is so vast. It acts as the 'brain' of the device. A weak microcontroller can pull down the overall performance of the entire system. With technology

evolving so fast connectivity with the external medium, power efficiency and good performance characteristics are very important when choosing a microcontroller.

4.2.1 The ESP-WROOM-32E Microcontroller

Espressif System has developed a very powerful and versatile module around the ESP32 board, named the ESP32-WROOM-32E. It is present on the variations of the board that specify in the name an ‘E’ at the ending of the name (e.g. ESP32-E). This letter also represents the fact that the board is a revision 3 model. Being revision 3 means that it is the last revision available on the market and it is more stable and reliable than pervious boards. This module incorporates a dual-core processor and multiple connectivity options, making it perfect for a wide range of applications such as Internet of Things (IoT) integration, small projects for hobbyists, testing components and so on. The dual-core 32-bit processor of the ESP32-WROOM-32E, named the Xtensa LX6, its especially known due to its ability to stand frequencies of up to 240 MHz This processing capacity makes possible the efficient execution of multiple tasks at the same time and by doing it becomes a crucial part in IoT applications where quick responses and reliability to various environmental stimuli are a priority. Other key benefits of the ESP32-WROOM-32E are the extended connectivity possibilities, having support for both Wi-Fi and Bluetooth. The Wi-Fi 802.11 b/g/n facilitates in the fast connection to the wireless networks. The Bluetooth 4.2 and Bluetooth Low Energy (BLE) ensure compatibility for a wider range of mobile devices and sensors. These extended connectivity possibilities are essential for applications that need fast and efficient data transfer in between multiple devices [12]. In Figure 2 a visual representation of the microcontroller is presented.

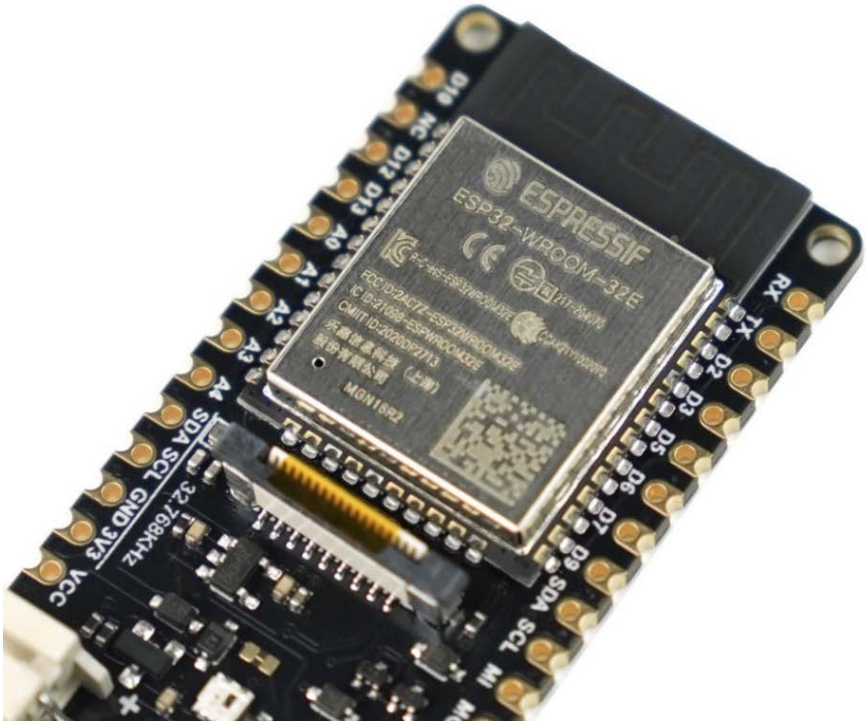


Figure 2. The ESP32-WROOM-32E Module [12]

ESP32-WROOM-32E is a powerful, generic Wi-Fi and Bluetooth Micro Controller Unit (MCU) module that targets a wide variety of applications. These applications may range from networks of low-power sensors to more demanding tasks such as MP3 decoding or voice encoding. The MCU comes with a PCB mounted antenna for the external connections. The on-board antennas save space by already being carefully positioned on the board and also provide signal integrity [13]. Inside the module as the core the ESP32-D0WD-V3, System on Chip (SoC), with the dual-core Xtensa LX6 that can be programmed to work with clock frequencies between 80MHz and 240 MHz This provides flexibility in performance and power consumption. Besides this, the users can choose to shut down the main CPU and rather use only the low-power co-processor in order to continuously monitor the peripherals. This function is extremely useful in applications where the energy is limited due to various factors. To extend the versatility of the ESP32-WROOM-32E MCU, we have to take a look at the interfaces that the module is compatible with. It includes analog pins, digital pins, UART interfaces, I2C interface, SPI interface and DAC interfaces [13]. These connections can be seen in Figure 3.

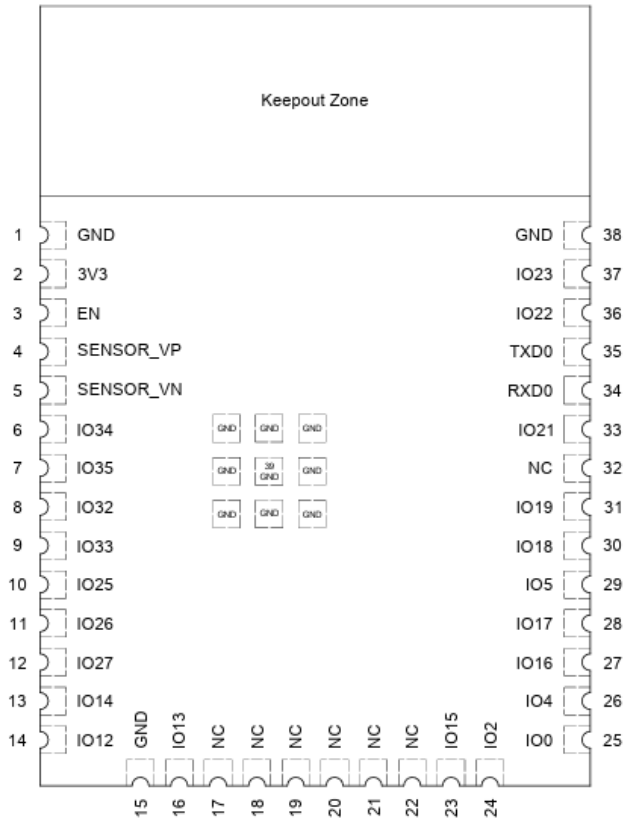


Figure 3. Pin Layout (Top View) [13]

This wide range of connectivity possibilities, the Wi-Fi, Bluetooth and BLE make this module future-proof and ideal for a big spectrum of applications. While the Wi-Fi enables a fast internet connection with other routers, with a long range and data transfer speeds of up to 150Mbps, the Bluetooth and BLE allows for the connection with mobile phones or low-power beacons. Both of these helping with remote monitoring of applications [13].

4.2.2 Functionality of the module

The dual-core microprocessor with low power consumption is built as a SoC. While the traditional take was to have an electronic device with a Graphics Processing Unit (GPU), Central Processing Unit (CPU), a dedicated signal processor, memory and other peripherals, the System on Chip take a different take. As microelectronics evolve the demand for smaller and more powerful chips has increased substantially. These SoCs combine all the previously mentioned components into one microchip with the only drawback of having to add a secondary memory if the system requires so. Advanced SoCs may include digital, analog and even mixed-signal or radio frequency capabilities when it comes to signal processing. These systems work as a complete computer system and can perform tasks quickly while maintaining a small size and low power and weight. They are often found in the automotive field, aircraft avionics systems, emergency notification devices, electronic gaming devices, IoT and many more applications [14].

For this study to investigate the technology in the ESP32-WROOM-32E, the datasheet presents the following:

- 448KB of ROM memory for base functions and booting
- 520KB of SRAM memory for instructions and data
- 16KB of SRAM memory in RTC for data storage
- 2MB PSRAM memory ideal for storing large data buffers
- On-board PCB antenna
- 40MHz crystal oscillator

4.2.3 Electrical considerations

In order for a complete system to function properly the datasheet provides any user with important data for safety and protection of the electronic components. Not respecting the recommended maximum values of the datasheet may result in failures and damage to the entire system. In Table 1 the recommended operating conditions are presented:

Table 1. Recommended operating conditions [13]

Symbol	Parameter	Min	Typ.	Max	Unit
VDD33	Power supply voltage	3.0	3.3	3.6	V
I_{VDD}	Current delivered by external power supply	0.5	-	-	A
T	Operating ambient temperature	85°C version	-40	85	°C
		105°C version		105	

The maximum ratings are also given and presented in Table 2:

Table 2. Absolute maximum rating [13]

Symbol	Parameter	Min	Max	Unit
$_{VDD33}$	Power supply voltage	-0.3	3.6	V
I_{output}	Cumulative IO output current	-	1.100	mA
T_{store}	Storage temperature	-40	105	°C

A common misunderstanding is how can the ESP32 have only 500mA of current delivered and also give a cumulative output current of 1100 mA. The big output is supposed to be the sum of all output current across all the GPIO (General Purpose Input/Output) pins. This current couldn't be delivered all at once due to power supply and thermal limitations. If the external power supply only gives in a input of 500 mA, then that is the current that must be shared between the pins. In the case of driving multiple pins at high currents, this can cause unwanted heat on the chip and potential damage to the hardware itself. From the datasheet, it is noted that for the DC characteristics a pin should be used with a current of 28 mA for low-level sinking or a 40 mA current in the high-level state. These values can be controlled by using external resistors in order to pull-up or pull-down the current or with the internal resistances of the board, but this needs to be specified in the coding of the MCU.

4.2.4 Comparison between microcontrollers

To take a look in why choosing a microcontroller board from the ESP family a table of comparison between different types of such modules was made. This table (see Table 3) was created in order to facilitate when deciding the right component for any application.

Table 3. Table of comparison between different types of microcontrollers

Criteria	Arduino	STM32	ESP-32
Data acquisition efficiency	Moderate (basic ADC, I2C, SPI; limited channels)	Excellent (high-precision ADC, DMA, fast GPIO, multiple timers)	Good (integrated ADC, I2C, SPI, UART; real-time data handling)
Price	Low to Medium (~$3-20)	Medium to High (~$8-20+)	Low (~$4-10)
Memory (Flash/RAM)	Low (ATmega328P: 32 KB flash, 2 KB SRAM; MKR: 256 KB flash, 32 KB SRAM)	Moderate (Up to 2 MB flash, varies by model)	High (Up to 16 MB)
Peripheral Availability	Basic to Moderate	Extensive (by adding external components)	Rich
Ease of use	High (Arduino IDE, huge community support)	Moderate (Advanced, more professional tools like STM32CubeIDE)	Moderate (Requires some setup, more technical)

The first one, the Arduino microcontroller (e.g. ATmega328p) used vastly differently for rapid prototyping and accessible project development comes as the least favorite. It is so because popular boards like the Uno or the Nano can be limited in terms of processing power, price being low to medium but in terms of space used in the actual project to use a better board it would take more physical space. For example, one of the most powerful boards in this category is the Arduino Mega that has a width of around 53 mm and a length of 101mm [15].

The second one, STM32, offers a processor that is designed to offer freedom to MCU users. As a package, the microcontroller unit offers very high performance, low power usage, real-time capabilities and low-voltage operation. The downside of this option is that the price ranges from moderate to high just for the chip without counting for the peripheral components and it requires a lot more precision when soldered to the board overall [16].

The third and final one, ESP-32 development board is equipped with the on-board antenna, Bluetooth, Wi-Fi, a dual-core processor with a clock frequency of up to 240MHz, up to 16 MB flash memory and has a rich peripheral availability (SPI, I2C, UART, ADC, DAC, PWM). It comes at a low price point, has good power efficiency and has large documentation. It is designed to provide good performance for applications such as a thermal imaging camera where reliability is the key and also lets the developer room to always improve the system [13]. With the external connection via the Wi-Fi on-board antenna that uses the IEEE 802.11 b/g/n standard for wireless communications the board is open to a lot more applications than the other competitors.

4.3 Temperature sensors

Temperature is an important factor in almost every field nowadays. Sensors are used to measure the temperature in a lot of different fields and applications. Some examples would be in the health and safety of humans, especially in the medical field but also in the usual surroundings because even simple things such as higher temperatures can lead to exhaustion or dehydration. Other applications where temperature variations plays an important role are in the electronics equipment and machinery fields because most of the electronic devices around us operate at a certain temperature and some of them even shut down if they are exposed to head. This problem can be also noticed in the energy efficiency domain where cooling systems and controlled heating can lead to reduced energy consumption and overall costs of a complex system [17].

4.3.1 Types of temperature sensors

1) Thermocouple

The thermocouple (Figure 4) is a type of temperature sensor that consists of two dissimilar types of metals, which are joined together at only one end. When the junction formed in between the two metals is either heated or cooled, it creates a voltage that can be calculated (Seeback effect). One disadvantage of this sensor is that the output voltage comes as a small value. In order to get good results with a thermocouple, precise amplification is needed and also protection to external noise over the wires. It has a temperature range of -200°C to 2300°C, comes at affordable pricing with moderate accuracy and low sensitivity in measurements [18].

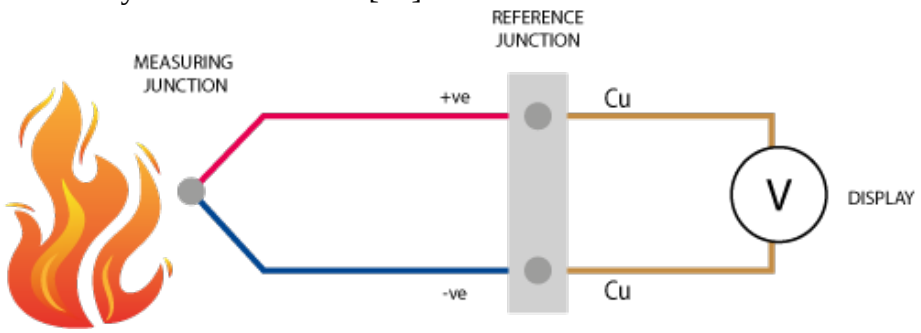


Figure 4. Example of a thermocouple [19]

2) The thermistor

The name of “thermistor” comes from term thermally sensitive resistor. These sensors are cost-effective and accurate in measuring temperatures with a high level of stability. They are also highly dependent on the material from which they are made. Some of the main materials used in making such a sensor are metallic oxides. Because they react like a resistor, the most used thermistors are those whose resistance increases with the temperatures decrease. They are mostly used in the -50°C to 250°C range [20][21].

3)The RTD

The resistive temperature sensor (RTD) is a type of temperature sensor based on some materials property that they increase resistance with the increase of temperature. By doing so, they are also called positive to temperature coefficient sensors (PTC). They are very precise sensors used in the -200°C to 850°C range. One of the most used materials for this sensor is platinum due to its almost linear response to the increases of temperature. It offers good accuracy and high sensitivity [21]. They are almost mistaken with the negative temperature coefficient (NTC) thermistors because they both vary their resistance with the temperature. In the figure below (Figure 5) a graph with both sensors characteristics is presented. The blue line being the characteristic of the RTD and the red line the characteristic of the PTC when it comes to resistance variations to temperature.

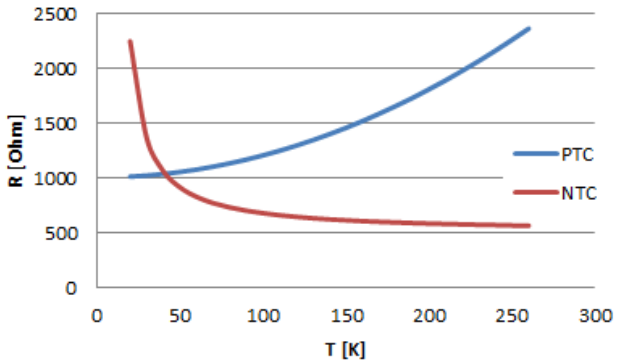


Figure 5. Graphical representation of how the resistance of NTC (red) and PTC (blue) vary with temperature [22]

4) Semiconductor-based sensors

Also known as the IC sensors (Figure 6) are a class of temperature sensors with an almost linear output, less accurate and operate best in the -55°C to 150°C range. They consist of two similar diodes or transistors encased in a chip. Therefore, they have a slower response time and also as a disadvantage they come at a moderate to high price point [21].

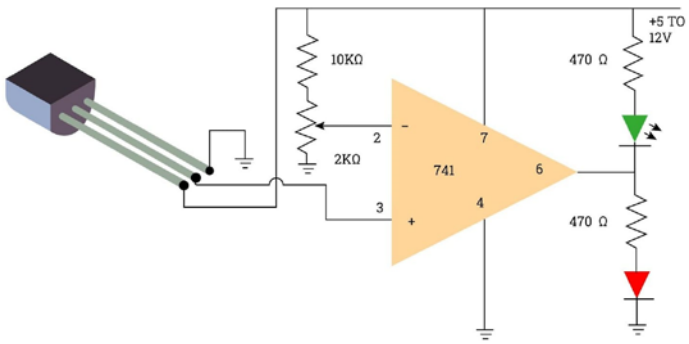


Figure 6. Inside of a semiconductor-based sensor [23]

5) IR thermal sensors

The infrared thermal sensing technology is a contactless way of obtaining accurate temperature readings in many applications where physical access may be out of reach or in industries where contact is prohibited. In the general maintenance field, this type of sensor is one of the best options because of its non-contact way of reading thermal data. They work by sensing the infrared radiation of objects and consist of a lens that focuses the IR energy on a detector which later converts that energy into an electrical signal. This data can be displayed as a thermal map or directly units of temperature [24].

4.3.2 Thermal imaging sensors

After we covered how thermal sensors work, its important to take a look at thermal imaging sensors. The common types described in the previous sub-sub chapter, the thermocouple, the thermistor, the RTD and the semiconductor-based sensor, each, use different methods and materials to sense variations of temperature [25]. When looking at thermal imaging sensors and thermography we can actually get to see these changes in real-time on a color display. The process of taking a thermal image is a rather sophisticated process but also non-invasive technique. This methos uses the infrared technology which converts the invisible, to the human eye, energy and converts it into a visible portrait.

Thermal imaging is a powerful and versatile technology that has become indispensable in many applications across multiple industries. By using a thermal imaging sensor, we can detect anomalies and changes in temperature. The most popular work by scanning their field of view and measure the temperature at multiple points or pixels. Each pixel having a small infrared thermometer, with the purpose of creating a grid with the temperature readings which can be visualized by the user. In the figure below (Figure 7) an example of a thermal image is presented from the human viewer's point of view.

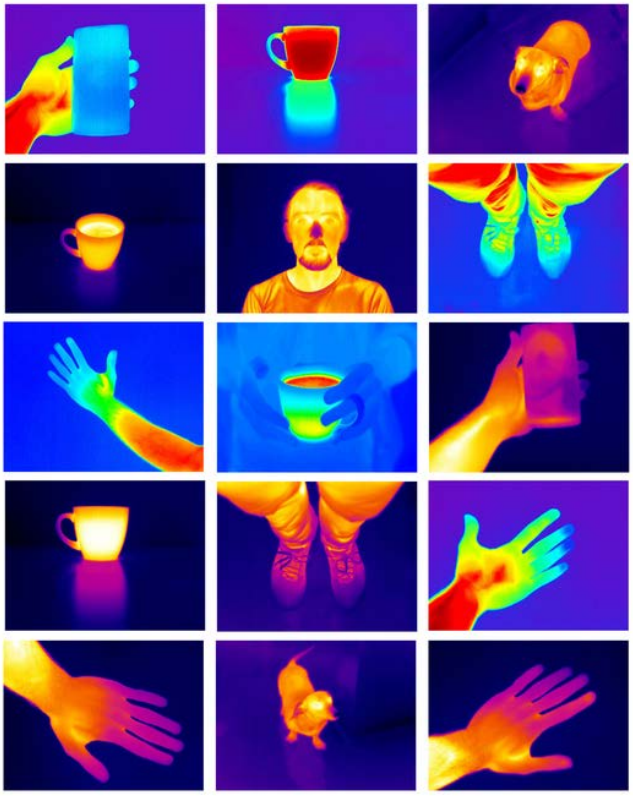


Figure 7. Thermal imaging versatility example [26]

A final temperature reading formula can be simplified as equation (2):

$$T_{object} = \sqrt[4]{\frac{IR_{Compensated}}{\varepsilon * \alpha}} + Ta_{ref} \quad (2)$$

After analyzing this image, we can observe how with the help of thermal imaging we can detect surrounding heat radiation of an object or even take that specific subject separately for further troubleshooting.

The market for thermal imaging modules is quite vast but only some stand out. They range from modules with smaller or higher resolution, from narrow fields of view to wider ones and so on. Of course, these characteristics come with a price. The technology behind such a module is very complex incorporating regulators, storage memories, acquisition systems and specific communication protocols for each system. These increase or decrease the overall quality of the camera's temperature range, accuracy, resolution and durability.

4.3.3 I2C protocol

For the digital interface of the device, it uses the I2C communication protocol with the support of FM+ mode, which means a clock frequency of up to 1MHz. The SDA and SCL (data ports) tolerate 5V so that the sensor can be connected to a I2C network of 5V. The communication session is initialized by a START condition and ends at the STOP condition. Each transition from HIGH state to LOW state in the SDA transmission line can generate the START condition and each transition from LOW to HIGH in the same transmission line can generate the STOP condition. Any of these actions need to be happening while the SCL transmission line is in the HIGH state [27]. The device addressing happens when the master sends a 7-bit address to the slave. The 8th bit is a Read/Write bit that indicates the direction of the transfer in the following manner:

- When in a HIGH state the master can read data from the slave
- When in a LOW state the master is able to send data to the slave

The acknowledgement happens during the 9th clock. If the receiver recognizes receiving a byte by pulling SDA line to a LOW state and does not acknowledge when it lets the SDA line in HIGH. Both operations are presented in Figure 8, respectively Figure 9.

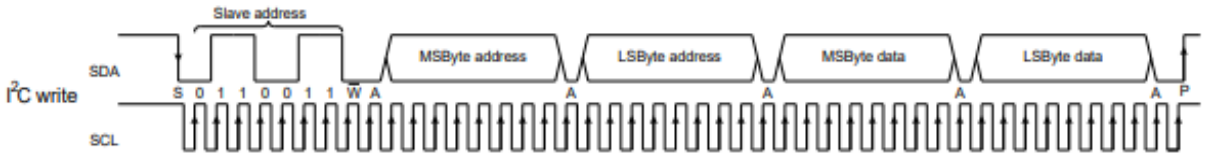


Figure 8. I2C write command format [27]

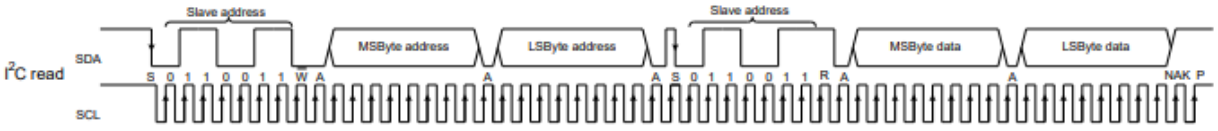


Figure 9. I2C read command [27]

4.4 The display

In embedded systems but also in electronic instrumentation, a display serves as the primary interface between the device and the user. They are used to provide real time and visual feedback. Liquid crystal displays (LCDs) are one of the most common class of displays because of their wide availability, versatility and low power consumption. The work by having a backlight which lights up the pixels electronically. Using liquid crystals, the LCD can rotate the polarized light. Such a display incorporates a polarizing glass filter that is located on both the back and the front of the pixels the liquid crystals [28].

These LCDs can be made with a passive matrix of pixels or an active matrix of pixels as the display grid. The passive matrix one has a grid of conductive elements with pixels at every intersection in the grid. The active matrix one is also called a thin film transistor display (TFT). This one requires less current to control the luminance because each pixel intersection has a transistor. This also helps with improving the screens' refresh rate [28].

4.4.1 The TFT display

. The TFT technology can offer an improved image quality when it comes to color displays and that makes them popular in applications where the visualization of data is mandatory. Most displays today can show color and in order to do so each color pixel needs to have three subpixels with filters of the colors red, green and blue [29].

As for the internal structure of the thin fil transistor display, it's best to split it in five sections and analyze each one:

- A liquid crystal layer: this one stands at the core of a TFT LCD, between two glass plates, containing molecules that are able to align in multiple directions when stimulated by an electric field
- TFTs: thin film transistors that function as switches that turn on when voltage stimulates them, allowing for the pixels to be lit up
- Pixel Electrodes: used to help with how the liquid crystals are oriented when a voltage is applied to them
- Color Filter: essential in the structure to control the colors of each pixel and enables LCD access to the whole color spectrum
- Backlight: used as a light source for the liquid crystals enabling the user to see images and colors

The working mechanism of the LCD displays with TFT technology is based on the concept of field-effect transistors (FET). The transistors terminals are the: source, gate and drain. Out of these three the FET transistors use the electrical field in order to control the flow of current. By doing this, a voltage is applied on to the gate of the transistor, allowing or denying conductivity between the source and the drain [30]. In Figure 10 the mechanism behind the TFT LCD is presented.

The contrast ratio of such display is calculated with the following formula (3):

$$Contrast\ Ratio = \frac{Average\ Surface\ Luminance\ with\ all\ white\ pixels}{Average\ Surface\ Luminance\ with\ all\ black\ pixels} \quad (3)$$

A high contrast ratio means that the results will have a sharper look, with better color definition and distinction. But for this application a gradient is also helpful when detecting temperature

modifications of a subject. This contrast ratio can also be affected by the viewing angle from which the observers look at the display [31].

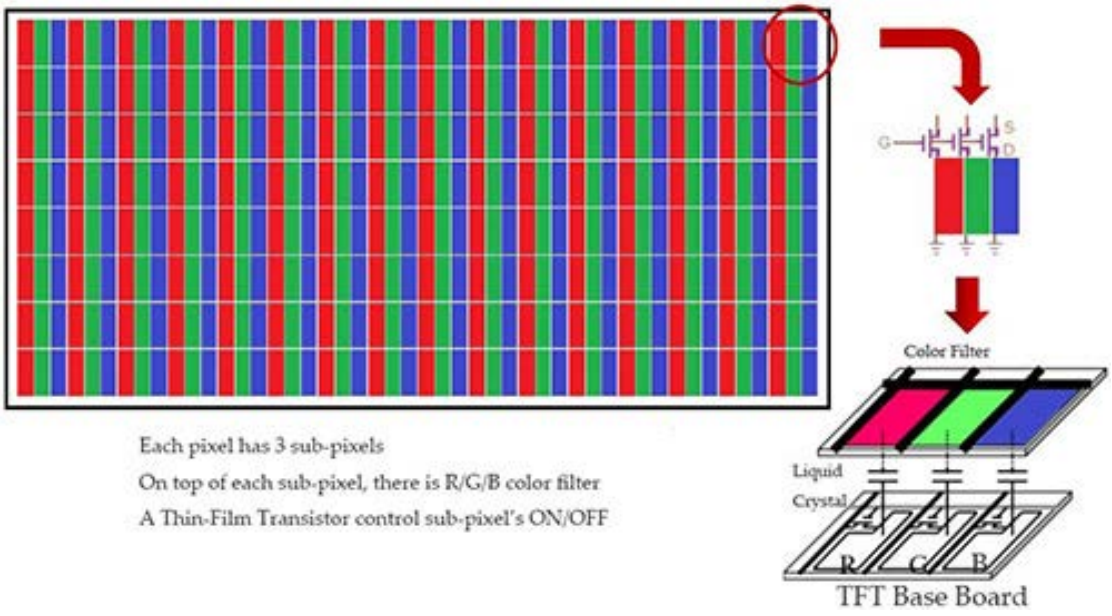


Figure 10. The working mechanism of the TFT LCDs [30]

4.4.2 SPI protocol

The serial peripheral interface (SPI) is a very popular communication protocol when it comes to interfaces between a microcontroller and surrounding peripherals such as sensors, acquisition systems and even displays. This protocol is synchronous, with full duplex interface. When a device generates a clock signal is called the main. Data transmission between this device and a sub node is always synchronized by the clock signal. The data is transmitted using the 'main out, sub node in' (MOSI) and 'main in, sub node out' (MISO) signals. This data is transmitted simultaneously. This communication interface lets the user to select if they want to shift the data and/or sample it on the falling or rising edge of the clock signal [32]. In Figure 11 a configuration example is presented for the SPI communication protocol.

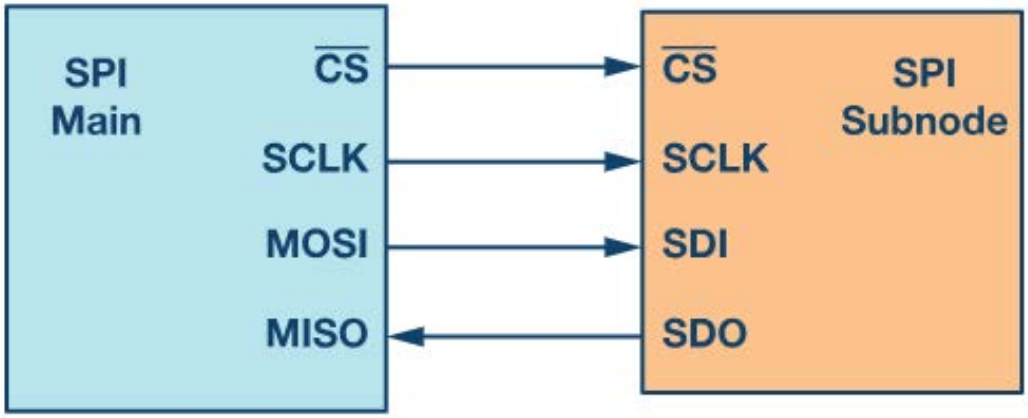


Figure 11. SPI configuration with main and a subnode [32].

4.5 The software

The software in every application is an essential part especially when working with microcontrollers. It can be described as a set of instructions that is used to execute specific tasks in order to operate computers of any kind. Also, as being the opposite of hardware, the term of ‘software’ is used to refer to scripts and programs that put a MCU to work. The general user of a device does not interact directly with the software. It just coordinates the hardware of a device handling the functions in the background and provides a digital platform for most of application [33].

4.5.1 Arduino IDE

The dedicated software for programming Arduino boards is the Arduino IDE. It is an open-source platform that was created in order to make the combination of hardware and software easier for any user. It allows the user to write code with a simplified version of the C and C++ programming languages. After the coding part is done, the script can be uploaded to an Arduino or ESP board. This program comes with its own integrated libraries that can be used for hardware components such as sensors, motors and communication modules. Because of its open-source characteristic, a lot of fans of this software have already created even more resources for anyone to use for free. These aspects make the Arduino IDE a great tool for anyone from beginners in electronics to even advanced developers [34].

5 Implementation

5.1 Block diagram

In the picture below (Figure 12) is presented the block scheme of the purposed system to be implemented. It shows how all the components interact with the chosen microcontroller. The center of the whole system is the ESP32-E, that is powered from a rechargeable battery which can be charged by using a TP4056 charging module. The data comes from a MLX90640 thermal imaging sensor and then displayed on a TFT LCD display. When the user wants the images displayed on the LCD can be sent to the SD card integrated in the system. All of these functional blocks from this diagram and the implementation of the system are detailed in the following paragraphs.

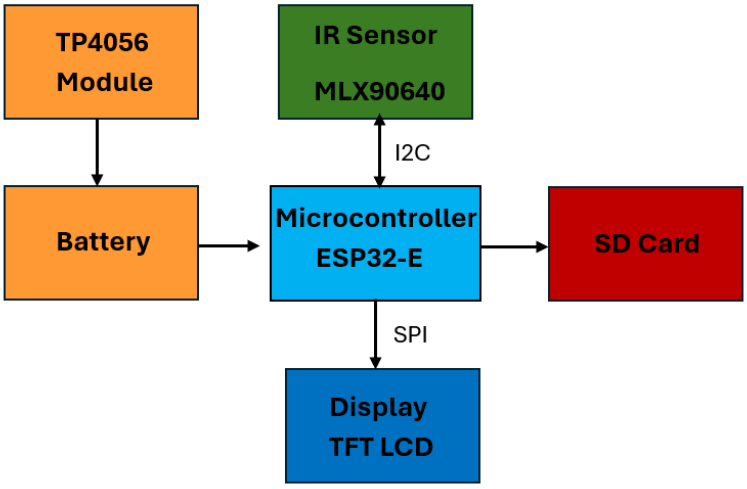


Figure 12. Block diagram

5.2 The power block

For the power part of the system, a charger module was chosen. The TP4056 is a very small module ideal for projects, and it is designed to give the efficient and safe charging of a rechargeable battery (Figure 13). This setup provides the portability of the whole device, also making it easy to be used in remote applications. As for the functional flow of the charging block, when the system is connected to an external power source through the charging module, it starts charging the battery.

When the battery is fully charged the module stops charging in order to prevent overcharging. After the switch is turned on, the battery starts to power the rest of the circuit. By connecting a simple two pin switch button to the system to control the state of the system, the user decides whether they want to use it or turn it off for the battery to charge. When in the off state, the device can be charged safely by plugging in the module using a micro-USB charger. When the switch is pressed the device starts and the 3.7V battery can start powering the system.

A Lithium-Polymer battery was used because of its rechargeable feature, compact size and high capacity. It has an electric charge capacity of 500mAh which means that if the system is in constant usage it would take about 45 minutes to be drained completely. Actual battery life can depend on various factors such as the battery age and health, the temperature and voltage drops over time but this one should be more than enough to supply the device in order to get the wanted results.

In the following figure an electrical schematic of the module is presented. The ‘B+’ and ‘B-‘ pins represent the terminals of the battery and the ‘OUT+’ and ‘OUT-‘ need to be connected to the switch and to ground, respectively. The two LEDs on the left of the schematic indicate when the battery is charging, if it has finished charging or it still needs time for it to be fully charged.

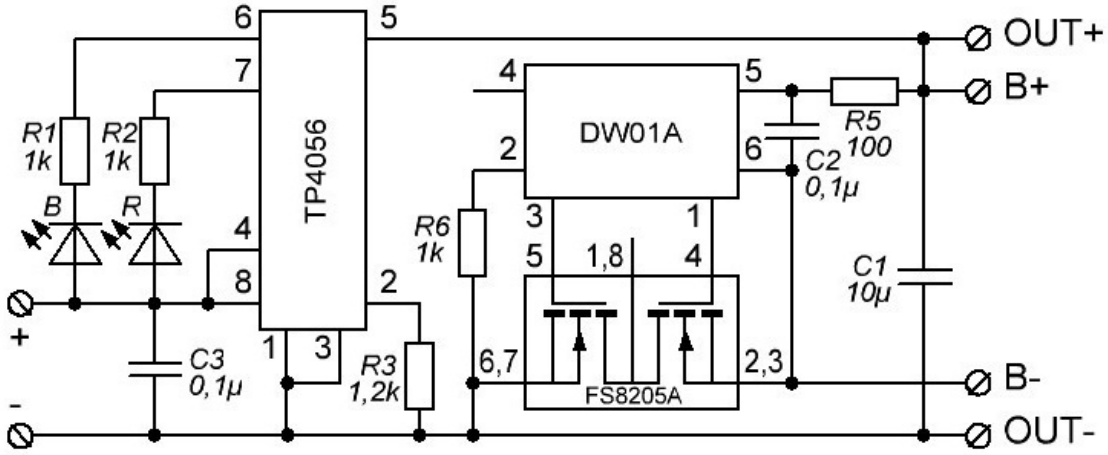


Figure 13. TP4056 Battery Charging Module with Protection Circuit [35]

5.3 The MLX90640 IR thermal camera

As an infrared radiation thermal sensing device, the MLX90640 stands out as one of the best in its field, being a small, reliable and cost-effective module. Any object with the temperature above the absolute zero (-273,15 °C) emits infrared radiation.



Figure 14. The MLX90640 module [27]

The module is presented in the figure above (see Figure 14). With the use of a 32x24 IR pixel array this sensor can measure the temperature of a subject in ca non-contact manner. It works even for a

long-distance measurement making use of the infrared technology. It can be easily integrated into many systems because if its size and ease of use. It comes in a TO39 package, which is commonly used in many sensors. Due to its compact size, it makes it easier to be integrated in a system. It is factory calibrated, assuring the user with a high precision in measurements. As a strong particularity of this sensor is the NETD (Noise Equivalent Temperature Difference) of only 0.1 RMS (Root Mean Square) when a refresh rate of 1 Hz is applied [27].

To keep its accuracy and precision the NETD is determined by multiple factors such as the electrical noise of the sensor, which is represented by the unwanted variations of the signals from the electronical components of the sensor, the capacity to capture thermal radiation emitted by objects, the sensitivity of the IR detector and the optical characteristics of the used lens. Therefore, the 0.1 K RMS values to the refresh rate of 1 Hz is considered a very good result in thermal imaging.

5.3.1 Electrical characteristics

The module comes in two variants one having the field of view (FOV) of 55°x35° and the other with the field of view of 110°x75°. As for electrical considerations it needs a supply voltage of 3.3V and has a current consumption of 23mA. It can observe temperature changes in the range of -40°C to 300°C which is great for most applications (see Table 4) [27].

Table 4. Electrical specifications

Parameter	Symbol	Min.	Typ.	Max	Unit
Supply Voltage	V_{DD}	3	3.3	3.6	V
Supply Current	I_{DD}	15	20	25	mA
Operating temperature	T_{AMB}	-40	-	85	°C
Storage temperature	T_{ST}	-40	-	85	°C
Target temperature	T_T	-40	-	300	°C

For the internal structure in the figure below a block diagram of the MLX90640 is presented (see Figure 15). A bandgap reference and PTAT sensor is present, with a regulator for the digital part and a 34 MHz RC oscillator. The array of M pixels is amplified and then the analog to digital converters helps with the measurements. The module sends digital data from the RAM memory inside the sensor and sends it to the microcontroller using the I2C communication protocol. Data about the calibration of the sensor and the temperature readings are found in the EEPROM [4].

The pins of the module are the following:

- VIN: power pin for the module, the sensor needs 3V DC in order to operate but it has a voltage regulator in order to run safely for any power source from 3V to 5V.
- 3Vo: this represents the output voltage of the integrated regulator and supplies 3.3V
- GND: generic ground
- SCL: serial clock for communication (I2C)
- SDA: seral data for communication (I2C)

For the I2C logic pins, both have a 4.7k resistor for pullup.

The refresh rate of the module is configured by the “Control register 1” with the address 0x800D. As an example, a 4Hz (having the “Refresh rate control” equal to 011), means that every 250ms another subpage data is delivered to the RAM memory [27]. The refresh rate can be programmed and stored in the EEPROM, so it does not require it to be reinitialized every time the device is powered on. The

sensor works by alternating between two subpages. The data from both pages are used in order to obtain the ambient temperature (T_a).

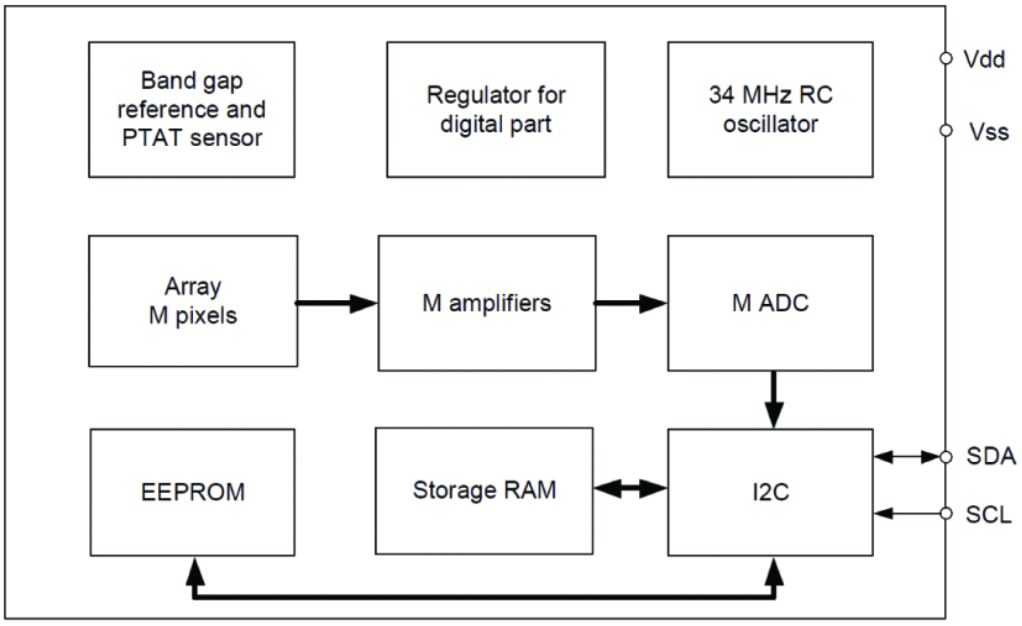


Figure 15. MLX90640 block diagram [4]

5.3.2 The setup of the MLX90640

The module is set to have a refresh rate of 2 frames per second, even if it can take up to 16 frames per second, for this application the thermal imaging camera was for more accurate than fast data readings. Also, this prevents potential noise that can appear at higher refresh rates. With the drawback of slightly increasing the processing time the ADC resolution is set to 18 bits. It is the highest resolution supported, and this improves the temperature readings accuracy. Even if the I2C interface supports communication at higher frequencies these characteristics were chosen so not a lot of noise can disturb the connections.

Where ε is the emissivity and α the pixel sensitivity coefficient. The emissivity refers to how a surface emits the infrared radiation, and the sensitivity ensures the accurate voltage to infrared intensity. To better understand how the MLX90640 sensor does the temperature calculation of a pixel from its array, the following calculations were made. Supposing that the subject's temperature is around 25°C, the ambient temperature 25°C (T_a), emissivity 1 (ε =1), sensitivity can be negated and taking the database from the sensor datasheet, the calculation flow of the temperature readings follows a set of rules:

- First a supply voltage calculation is made
- Then the ambient temperature calculations are made
- A gain compensation
- The offset, VDD and ambient temperature are compensated
- The infrared data gradient and emissivity compensations are calculated
- Sensitivity is normalized
- Final temperature value is given

When we add all the values, the sensitivity is divided to (1− ε) and when using equation (2) the final temperature of the object T_o gives us the exact measurement of 25°C. This algorithm is done for all the pixels in the 24x32 array of the module.

It can observe temperature changes in the range of -40°C to 300°C which is great for most applications. As for the resolution of the module, it uses a matrix of 768 FIR pixels. To denote each pixel the 'Pix(i, j)' expression is used where 'i' indicates the line number (from 1 to 24) and 'j' indicates the number of the column (from 1 to 32). The module reads the data in a checker-board pattern and the FOV of 110°x75° makes for the module to capture more subjects in a frame. The data is read twice per frame. This pattern can be observed in the image below (see Figure 16).

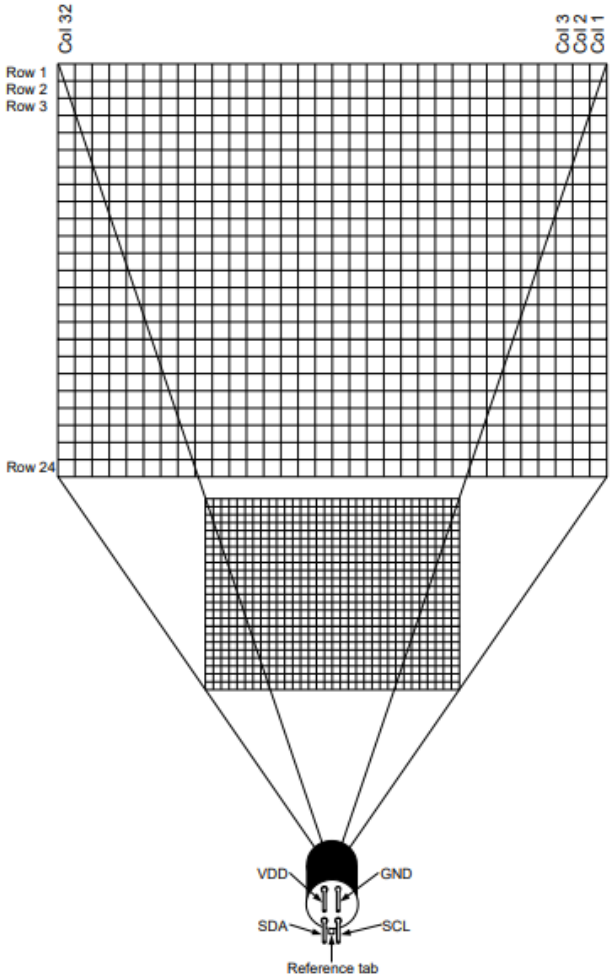


Figure 16. Pixel in the whole FOV [27]

5.4 The TFT LCD display

For the display of the device a 2.8" diagonal, TFT display was chosen, it has 4 white LEDs for backlight and a colorful 240x320 display resolution. The colors are RGB format with individual pixel control, giving the display a greater resolution than a classic black and white display. It comes with a

resistive touchscreen feature, but it was not used in this application. The display has two operational modes: one being the 8-bit display mode and the SPI mode. For the 8-bit display setting it uses eight digital data lines and additional control lines with a total of 12 lines. This mode was not used, and the pins are not connected. They are only used as mounting pins for the display to stand in the right position. As a slower method to read and write data on the display the SPI setting lets the developer use only 5 pins and it is easier to set up. This setting was used in this project. The display front and back are presented in the next figure (see Figure 17) [36].

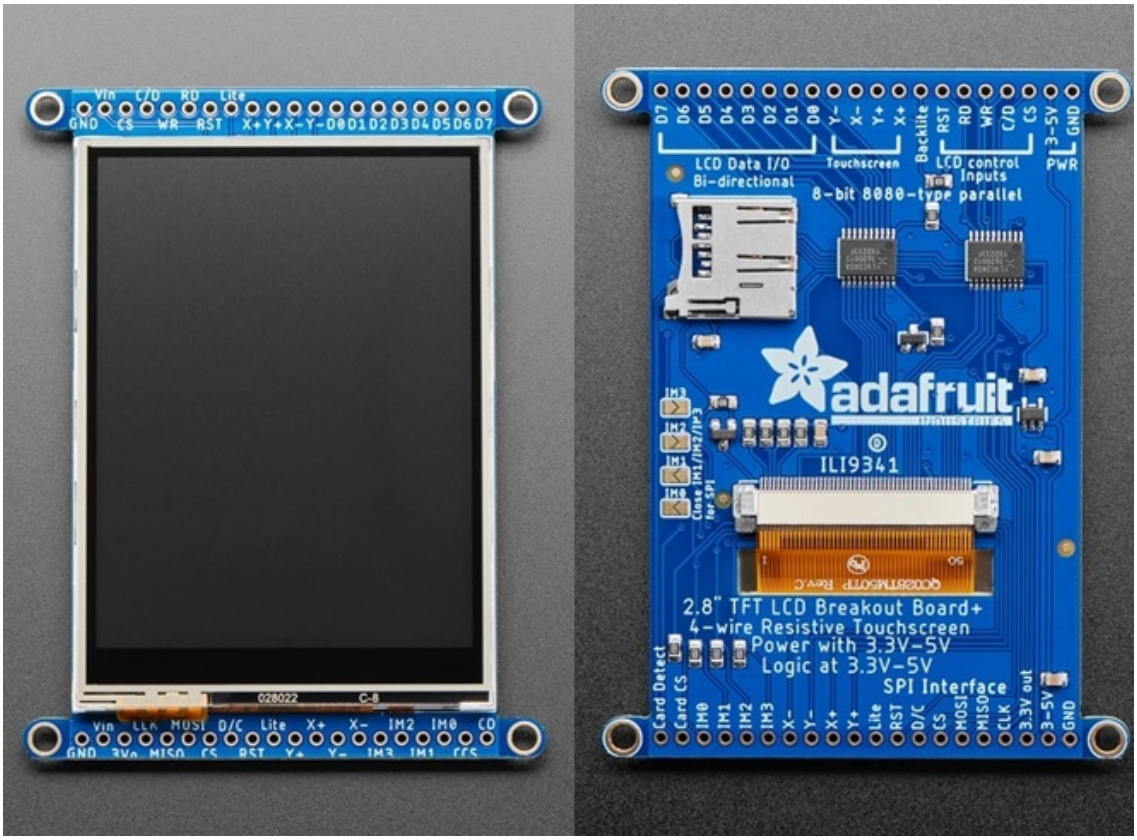


Figure 17. Display front (left-side) and back (right-side) for reference [36]

5.4.1 Electrical characteristics

For the electrical characteristics of the display the following table was made (Table 5). The DC characteristics indicate that the supply voltage needed for the display is around 2.8V but it can support 3.3V connection. It is also compatible with a 5V source because it has a 3.3V and 150 mA regulator integrated.

Table 5. DC characteristics of the display

Parameter	Symbol	Min	Typ.	Max	Unit
Power supply voltage	VCI	2.5	2.8	3.3	V
Supply current	IDD	80	-	150	mA
Input voltage ‘H’ level	VIH	0.7VDDI	-	VDDI	V
Input voltage ‘L’ level	VIL	VSS	-	0.3VDDI	V

5.4.2 The setup of the TFT display

The ILI 9341 controller is made as a SoC driver for TFT displays. It is specially made for LCDs with 240RGBx320 dots displays. This driver can handle eight-, nine-, sixteen- or eighteen-bit data bus in parallel with an MCU interface, six-, sixteen- or eighteen-bit data bus with the RGB interface and three or four lines with the serial peripheral interface (SPI). As for the electrical point of view it can operate with 3.3V [37].

For the SPI communication protocol, only 7 pins in total are required to be used:

- The SPI data in (MISO)
- The SPI data out (MOSI)
- chip select (CS)
- Data/command (D/C)
- clock (CLK)
- generic ground (GND)
- supply (3-5V Vin)

Also, for this protocol to function, three jumpers have to be soldered. They are found on the back of the display and labelled IM1', IM2' and 'IM3'. By doing so the SPI communication mode is selected [38]. This protocol is synchronous, with full duplex interface. When a device generates a clock signal is called the main. Data transmission between this device and a sub node is always synchronized by the clock signal. The data is transmitted using the 'main out, sub node in' (MOSI) and 'main in, sub node out' (MISO) signals. This data is transmitted simultaneously. This communication interface lets the user to select if they want to shift the data and/or sample it on the falling or rising edge of the clock signal [32]. The chip-select pin is programmed in order to enable or disable the ILI9341 controller. It also has the possibility of an external reset with the help of the RST pin. A write cycle sequence happens when the WRX signal switches from high state to low state and then it is pulled back the high state during the write cycle. The information is displayed from the host processor only on the rising edge of the WRX signal. The read cycle sequence state switching happens just like the write sequence and is also read on the rising edge of the RDX signal [37].

For this setup of the display using the SPI protocol, in the figure below it is presented how the color is formed in the 4-line serial interface (Figure 18).

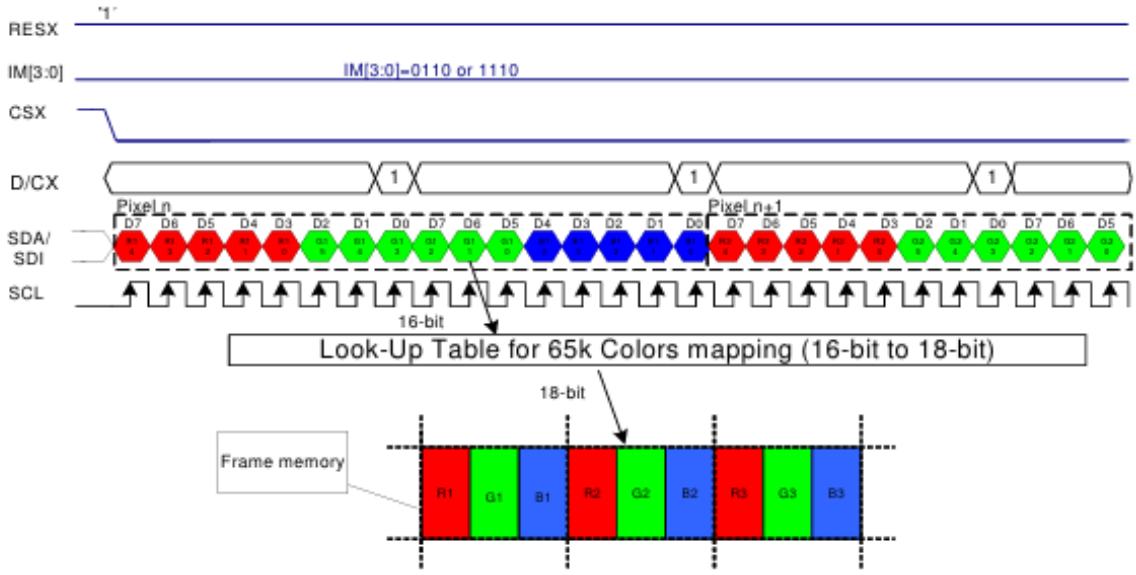


Figure 18. Color detection of each pixel using the SPI interface [25]

By using the SPI interface in the connection of the display to the rest of the system, we can also benefit from the SD Card slot connected on the back of the display (see Figure 17, right side). It was used in order to get the current frame from the display sent to the SD card. This feature is very helpful if we want to get the data and analyze it later. A 32 GB memory card was chosen for this.

5.4 The ESP32-E microcontroller

The chosen microcontroller for this implementation was the FireBeetle 2 ESP32-E (Figure 19) with the extended flash memory of 16 MB and PSRAM of 2 MB. This decision was made because this version is the latest one, fact known from the ‘E’ in the name which represents a latest revision sample. Also, the FireBeetle series is known for being used in battery-powered applications because of its economical energy usage. By configuring it correctly, and with the right peripherals, it can run continuously. This makes it perfect for the LiPo battery chosen.

It is also renowned for Bluetooth and Wi-Fi connectivity, but as if for now these features were not used, making future improvements possible. In its structure we notice the Espressif’s ESP32-WROOM-32E-N16R2 module. All these put this microcontroller board stand out and put it on top of its competitors [12].



Figure 19. The FireBeetle 2 ESP32-E (N16R2) Microcontroller [12]

By choosing this module we can take advantage of the 11 analog pins, 17 digital pins, 1 SPI interface, 1 I2S interface, I2C interface, 2 DAC interfaces and 3 UART interfaces for a lot of connectivity possibilities. All of these while maintaining a small size of 25.4mm by 60 mm [12].

5.4.1 The setup of the ESP32-E microcontroller

For the setup of this microcontroller in this project, the following connections were made for the temperature sensing module:

- The 3.3V from the ESP was connected to the 3.3V supply
- The ground is connected to GND
- The SDA of the ESP was connected to the SDA pin
- The SCL of the ESP was connected to the SCL pin

Even if the sensor module can be supplied from 5V it is no need to take advantage of the low-dropout regulator (LDO) of the MLX90640 because the 3.3V pin of the ESP is enough.

For the setup of the TFT LCD display, more connections were made in order to use the SPI connection, SD card access, and control of the overall display. The connections are as it follows:

- A 3.3V connection to the supply of the display
- The ground is connected to GND
- The SCK of the ESP is linked to CLK (SPI)
- The MOSI of the ESP is connected to the MOSI (SPI)
- The MISO of the ESP is connected to the MISO (SPI)
- The GPIO2 pin of the board is linked to the CS (SPI)
- The GPIO13 pin of the board is linked to the D/C
- The GPIO14 pin of the board is linked for the RST (reset of the display)
- The GPIO0 is used for the backlight control (Lite)
- The GPIO26 is used to access the SD Card from the display board (SD_CS)

It the same way it was done for the sensor, the supply is connected directly to the 3.3V pin of the LCD, not needing to use the on-board mounted LDO of the display.

In order to make use of the SD Card, after the initialization was done, a button was placed in the system. The button is connected to the GPIO 17 pin of the ESP-32E and to the internal pull-up resistor. The pull-up resistor connects the pin to 3.3V and the pin reads a ‘HIGH’ state. When it is pressed, the pin reads a ‘LOW’ state, or a direct connection to ground. This method was used so no other external resistor is needed and the debouncing is easier to be integrated in the software. For each jump from ‘HIGH’ to ‘LOW’ state the microcontroller can send a copy of the current displayed image to the SD Card. It is also prevented with a delay of 500ms so the button cannot be pressed multiple times and it gives enough time for the pictures to be sent to the external storage.

The SD Card chosen is a 32GB card formatted as FAT32 which is the default system for the ESP32-E. A bigger card cannot be supported by the microcontroller. This format is great for fast initialization and consumes less of the RAM memory and also consumes less current. The pictures are on a BMP format because they don’t need any additional compression.

5.5 The PCB

The PCB was designed in KiCad which is a free tool used for a variety of projects. It lets the developer create its own schematic of the design, choose the wanted footprints and create the desired layout. It also lets the developer see the design by having its own 3D viewer of the PCB. Of course, the program comes with the help of the designer with verification tools such as the Electrical Rule Checker (ERC) for the schematic design and the Design Rule Checker (DRC) for the PCB layout. The Gerber files of the PCB were sent to an external manufacturing company for fabrication.

As an example of a messy and unorganized design the following figure the testing on a breadboard and the needed wires is presented (Figure 20).

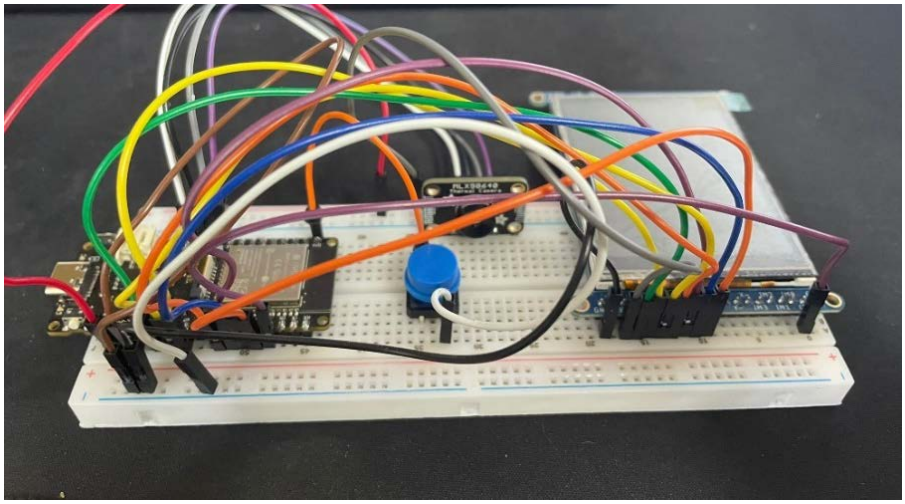


Figure 20. Test implementation using a breadboard and wires for connections

This configuration was made in order to test and validate the circuit. Problems like wrong voltage levels and connection configurations mistakes. All of these components are connected with each other with a custom-made printed circuit board (PCB). This decision was made in order for the device to have a better connection plan overall look like a final product.

As for the mechanical aspect, a PCB will always keep the components still in the right place. The purpose of being just for the connections is the most reliable option for a device. Of course, it comes with a price, but it was made on a 2-layer design, so the cost is lower than any of the components used. Also, connectivity would no longer be a problem with the custom-designed board. It is made with 4 mounting holes so it can be easily attached to a casing. This makes the whole device measure in size 120x70 mm. The final form of the PCB is presented in the following figure (Figure 21). This provides a better connection for high-speed signals such as the I2C, and SPI used in making this project. Also, this makes for a more compact layout and organized design.

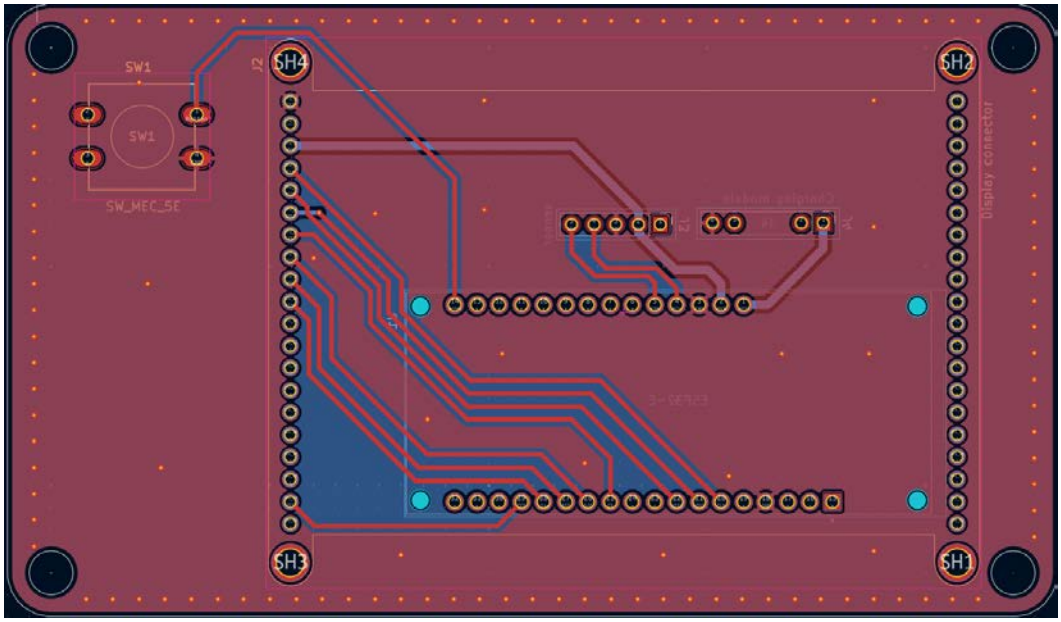


Figure 21. Front view of the PCB design

The connection tracks were routed all on the top layer of the PCB and underneath the display, because it takes up almost all the space of the design. All of them are of 0.5mm width, ensuring good connection. No vias were used except for the MOSI track which needed a crossover at the ending. The button was placed in the top right corner of the design, so the track connected to it is as short as possible. From the bottom side of the ESP footprint, a separation can be observed. It was done in order to isolate the SPI data lines respecting general guidelines for high-speed interfaces. All around the PCB on the edges stitching vias, connected to ground, were placed to improve the electrical performance. Also, more vias were placed to ensure the GND layer is continuous and to connect the top and bottom layer. Rounder corners were made for a cleaner look and for each corner a mounting whole was placed for the design to be easily encased if wanted.

On the bottom layer power signals were routed with a track width of 1mm. The rest of the board is filled with a ground plane. In the figure below an image of the bottom layer design is presented (Figure 22).

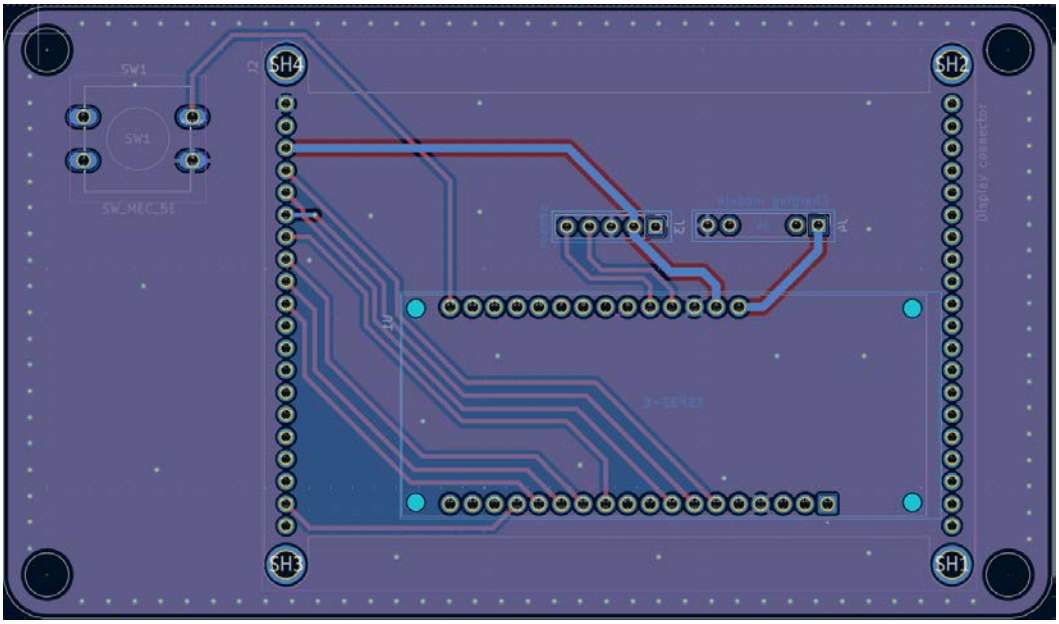


Figure 22. The bottom layer design of the PCB

On this layer the microcontroller, the thermal imaging sensor and the charging module were placed. All the components are of through hole type making routing on both layers accessible.

5.6 Software implementation

Arduino IDE is the program used for the software implementation. In order to be able to write the best code for this application multiple open-source libraries were used to make the development more easier. Each library used has a specific role:

- ‘Wire.h’ was used for the I2C communication of the MLX90640 module
- ‘Adafruit_MLX90640.h’ for the temperature calculation of the sensor
- ‘Adafruit_ILI9341.h’ was used for the TFT initialization and display of the thermal maps
- ‘Adafruit_GFX.h’ was added for the text display on the screen feature
- ‘SPI.h’ was used for the SPI communication of the LCD
- ‘SD.h’ was used fot the SD card initialization and save of the thermal images

All of these are used so the device can properly read the thermal data and display it on the LCD. With a software interrupt the image from the display can be sent to the SD Card. All of these steps were verified in the serial monitor of the IDE before implementing the final solution to check is all the connections are right. When using multiple hardware components, it is important to keep a clean workflow of the software. This system lets the user get data in real-time for the compact thermal imaging camera. The code runs continuously, and the process can only be ended when the device is shut down by the user. Each separate function in the code has its own part and order of execution and they will be discussed in the following sub-sub chapter. This program can always be improved because of its fluent flow, keeping the system perfect for future improvements. As a whole for a better understanding of the code, the following flowchart was made available in Figure 23.

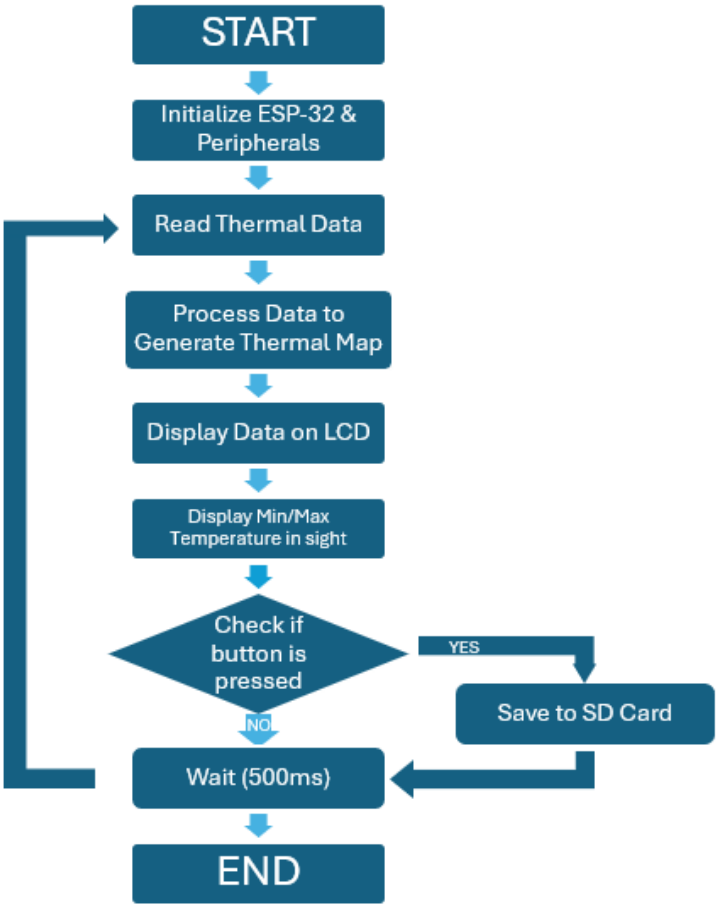


Figure 23. Flowchart of the code

5.6.1 Coding blocks

As mentioned before, each part of the flowchart has its own coding part. The more interesting ones are presented. These examples stay at the core of the functionality of the entire device. For example, in the picture below (Figure 24) after the pins were defined and the initialization of the microcontroller and its peripherals was made, the color mapping function is presented.

```
uint16_t getColor(float temp, float minTemp, float maxTemp) {
  float ratio = (temp - minTemp) / (maxTemp - minTemp);
  ratio = constrain(ratio, 0, 1);

  uint8_t r, g, b;
  if (ratio < 0.5) {
    g = (uint8_t)(255 * ratio * 2);
    b = 255 - g;
    r = 0;
  } else {
    r = (uint8_t)(255 * (ratio - 0.5) * 2);
    g = 255 - r;
    b = 0;
  }
  return tft.color565(r, g, b);
}
```

Figure 24. Color mapping function

This function sets the RGB color gradient on the display marking the blue color for cold subjects, green to yellow for medium temperatures that are near the ambient temperature and red for hot subjects.

For the saving on the SD card the images were enhanced 4 times for a bigger frame. The original pictures came with the resolution on 24x32 pixels, which was quite hard to see but with the use of this enhancement the 96x128 resolution images got much clearer. This code is presented in Figure 25.

```
bool saveFrameToSD(const float *frame, float minTemp, float maxTemp) {
  const int scale = 4;
  const int width = 32 * scale;  // 128
  const int height = 24 * scale; // 96
  const int rowSize = width * 3 + (width * 3) % 4; // padded to 4-byte alignment
  uint32_t fileSize = 54 + rowSize * height;
```

Figure 25. Enhancement of the image

The display can be adjusted to the liking of the developer with a simple function which changes the rotation of the actual image on the display. For doing so, in this application the ‘tft.setRotation(1)’ function was used. The button that is used also had to be adjusted in code. In order to not use any other external resistances, the internal pull-up resistor from the ESP-32E was used.

As for the MLX90640 sensor, 3 lines of code are the most important when initializing the module. They dictate the mode in which the sensor operates (chess-mode), the resolution of the sensor (18-bit resolution) and the refresh rate (2 Hz). This initialization is presented in Figure 26.

```
mlx.setMode(MLX90640_CHESS);
mlx.setResolution(MLX90640_ADC_18BIT);
mlx.setRefreshRate(MLX90640_2_HZ);
```

Figure 26. Initialization of the MLX90640

The main loop of the code holds the data for capturing the main frame, the calculation of the minimum and maximum temperatures, the orientation of the display the temperature to color mapping function, the code for the button press action, the delay and also the code for two boxes that are shown only on the display with the min/max temperatures detected. An example of this code is presented in Figure 27. This was the section for the display the box of the minimum temperature seen by the sensor.

```
tft.fillRect(flippedXMin, flippedYMin, boxWidth, boxHeight, ILI9341_BLACK);
tft.drawRect(flippedXMin, flippedYMin, boxWidth, boxHeight, ILI9341_WHITE);
snprintf(tempString, sizeof(tempString), "Min:%.1fC", minTemp);
tft.setTextColor(ILI9341_WHITE);
tft.setTextSize(1);
tft.setCursor(flippedXMin + 5, flippedYMin + 5);
tft.print(tempString);
```

Figure 27. Code for the addition of a box with the minimum temperature reading in sight

5.7 The casing

For the system mentioned above, a custom-made 3D printed case was used for housing all the components. It was designed to fit the device dimensions in the best way possible, making sure that the components have enough space in it, housing the PCB with the soldered components and also the power part which includes the battery. It has cutouts for SD card access, the ON/OFF button and grants access to the micro-USB port for the rechargeable battery.

This case is used to protect the internal hardware for dust or any external damage making visible to the user only the display, the button used for saving the images and the sensor for measurements. The casing can easily be modified and reprinted if needed for any future improvements to the system.

Below the 3D rendering of the casing is displayed:

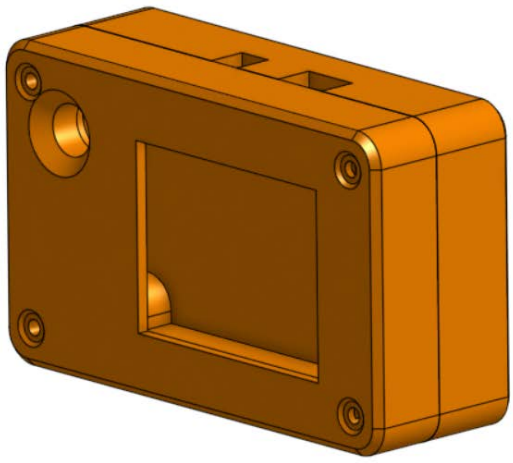


Figure 28. User view of the 3D printed enclosure

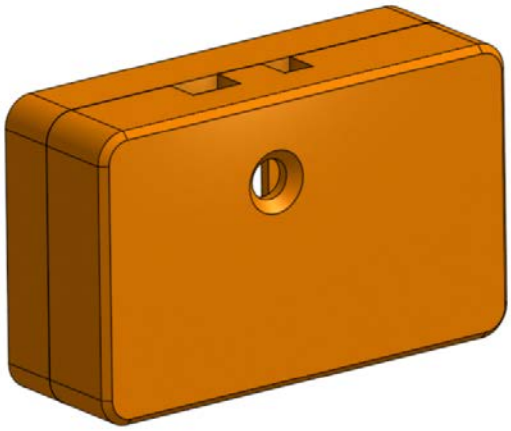


Figure 29. The back of the 3D printed enclosure

In Figure 28 is presented a better view of the user view of the device with a square cutout for the 2.8-inch display and a round cutout for the button used. On the two the cutouts are for the power button and the TP4056 module used for recharging the battery. The case is held together by screws in the corners connecting the front and the bottom of the enclosure with the mounting holes of the PCB offering stability when the device is handheld.

In Figure 29 only one hole was needed for the temperature measuring camera of the module and the rest is left sealed in order to protect the components.

6 Experimental results

In order to evaluate the performance of the device firstly it was assembled and put into the 3D printed enclosure mentioned in the last chapter. Of course, after this procedure was done the only thing left was to do experimental tests in different mediums and in different situations. The experiments had the purpose of determining the device's reaction in different environments and conditions and also to test the temperature reading accuracy, image clarity and the thermal sensitivity. By doing these tests we find the advantages and disadvantages of the device and the most important features are presented in this section, documented with data analysis and thermal images to provide the best documentation.

For the final view of the system in Figure 30 is presented the inside of the device where the ESP32-E microcontroller (1), the MLX90640 sensor (2), the TP4056 charging module (3) and the battery (4) can be seen. These are the internal hardware parts that the user dosent need to intereact with. To limit the access to these components they are placed behind the back view of the casing (see Figure 30) and the only thing that can be seen for the user point of view here is the camera from the thermal sensing module. The battery was also put on sponged on both sides in order to protect the PCB integrity in case of overheating. Some wires were used to connect the power part to the microcontroller but all of them cannot be seen by a user because they are sorted in such a way to fit with ease in the casing.

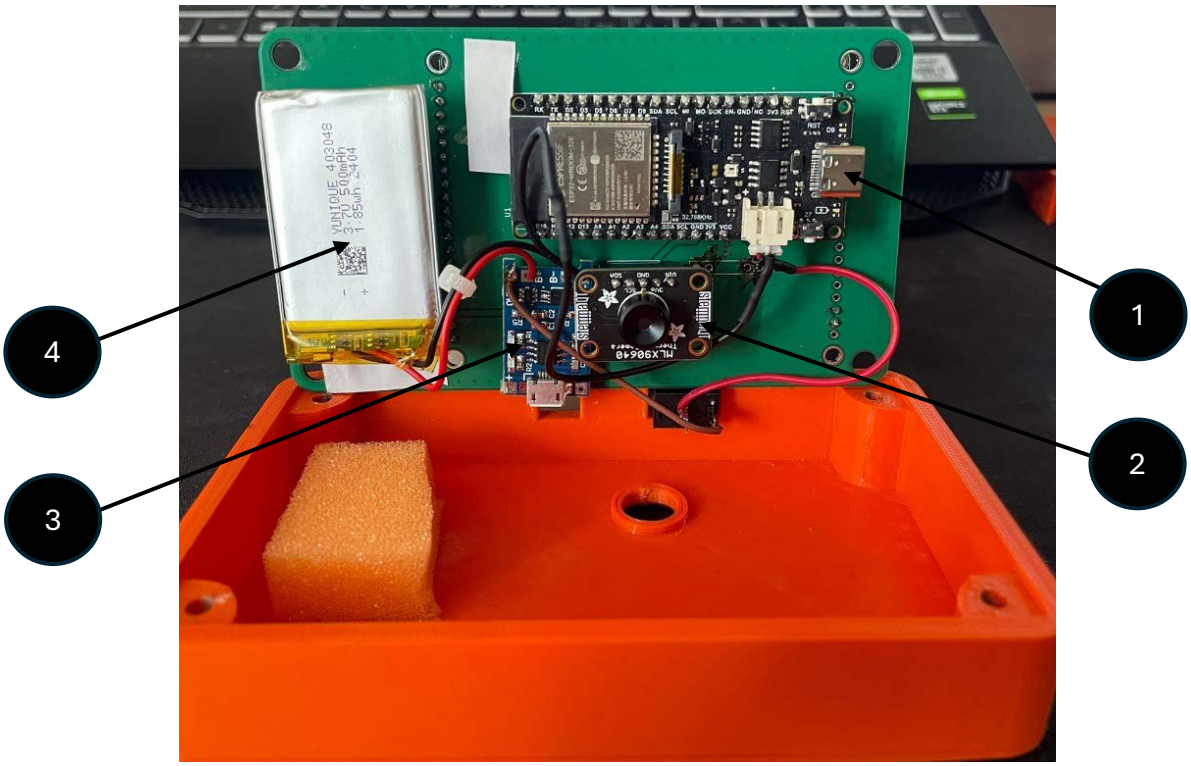


Figure 30. Final implementation of the hardware that is not visible to the user of the device

The part that the user sees behind the device is the LCD TFT display (1) and the button (2) used to save the images. They are presented in this paper in Figure 31. They are the only parts that matter to the average user. The case is held together by screws to keep it stable when it is put in a moving environment. Also, in this picture it can better be observed the ON/OFF switch and the cutout of the charging port.

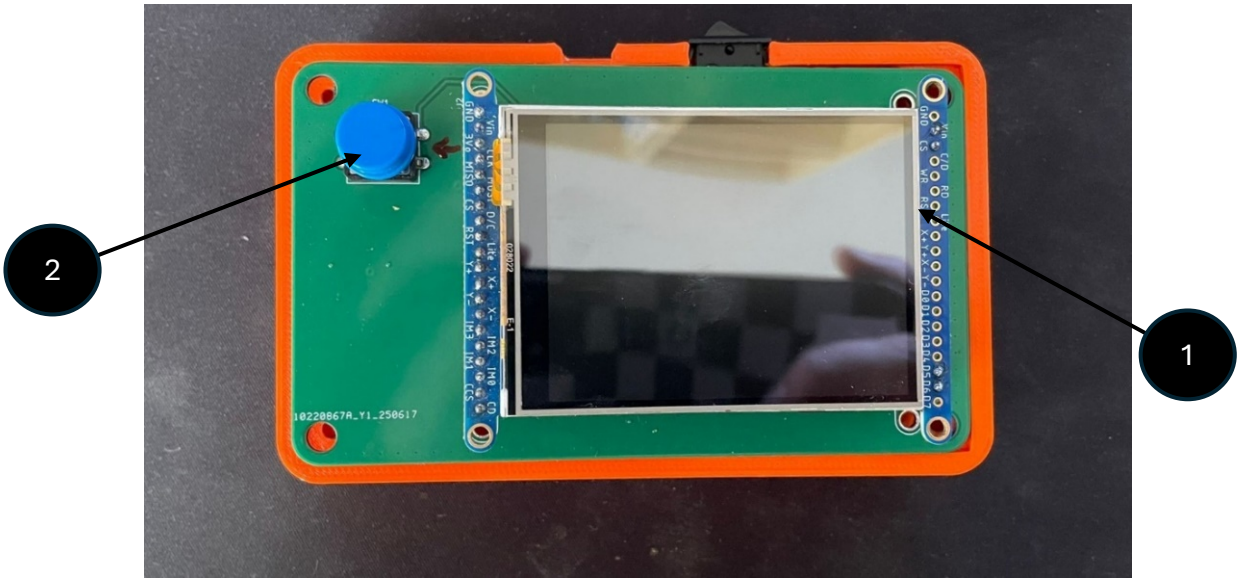


Figure 31. Final implementation of the components that are visible to the user of the device

6.1 Testing for electrical components heating

For the first test in this project was tested the behavior of a Raspberry Pi zero circuit board. This board is known to reach higher temperature values while in use due to its processing power. That's why the more advanced models have even coolers installed but the 'zero' version doesn't have one, so it was a perfect test for the device. In Figure 32 the testing is presented and in the next picture (Figure 33) the image saved from the test is presented.

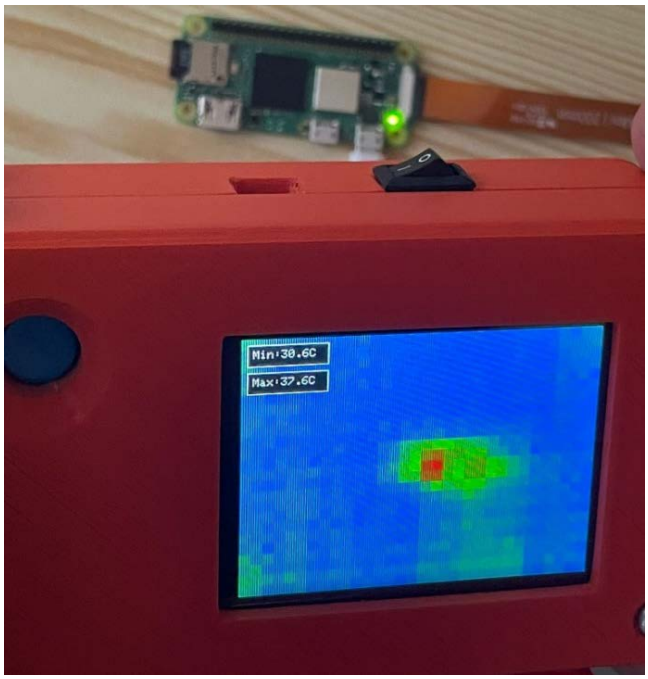


Figure 32. Temperature scanning of the Raspberry Pi zero

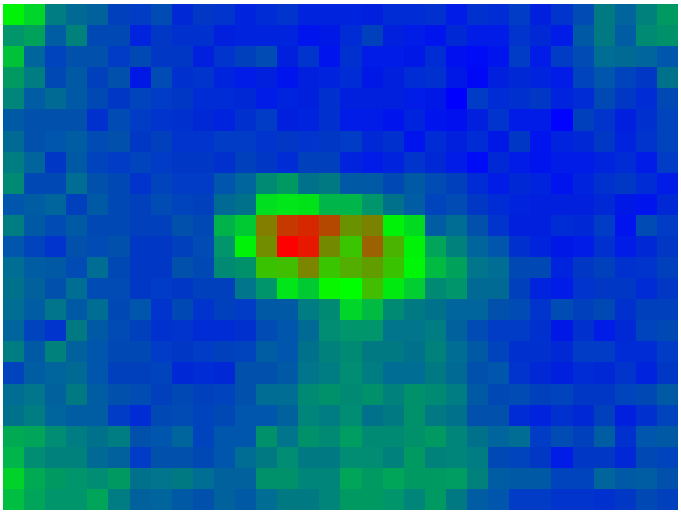


Figure 33. Thermal image taken with the device of the Raspberry Pi zero board

In this test it is observed that the Raspberry Pi has an area in the left side that heats up more than the rest of the board. The reading of the device used in this study says that the maximum temperature seen in sight is around 37.6°C.

6.2 Testing for negative temperatures

For the second test the device was set to measure a negative temperature. This was done by taking out a ice pack from the freezer and pointing the camera at it. The expected result is of course a negative temperature close to the one of the freezers. In the figure below (Figure 34) is presented how the testing was done.

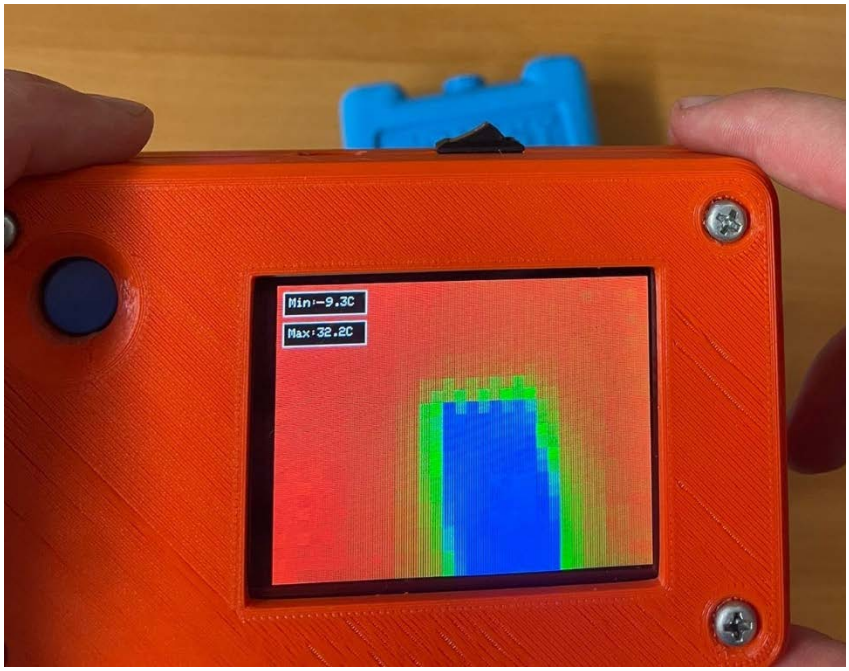


Figure 34. Negative temperature reading test

As expected, the result is a negative temperature. For this test we had to look at the minimum temperature that is shown to be around -9.3°C. In the next picture a image taken of the object with the thermal imaging device is presented (Figure 35).

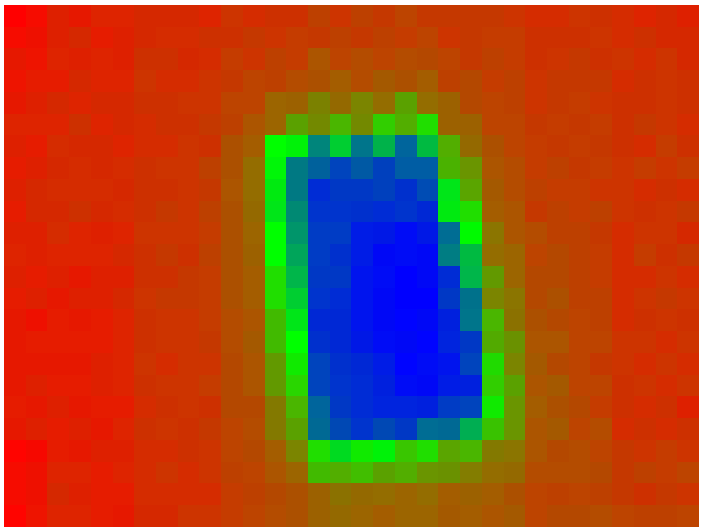


Figure 35. Thermal image taken with the device of the ice pack

6.3 Test for detecting a person

In the third test, the device was set to detect the temperature difference of a person in a room. This test is especially helpful in the safety and security domain the person being seen very clearly in with a outline compared to the ambient temperature of the room. The person stands at about one meter from the device. The distinction is made because the person has a higher temperature than the ambient. In Figure 36 the setup for this test is presented.

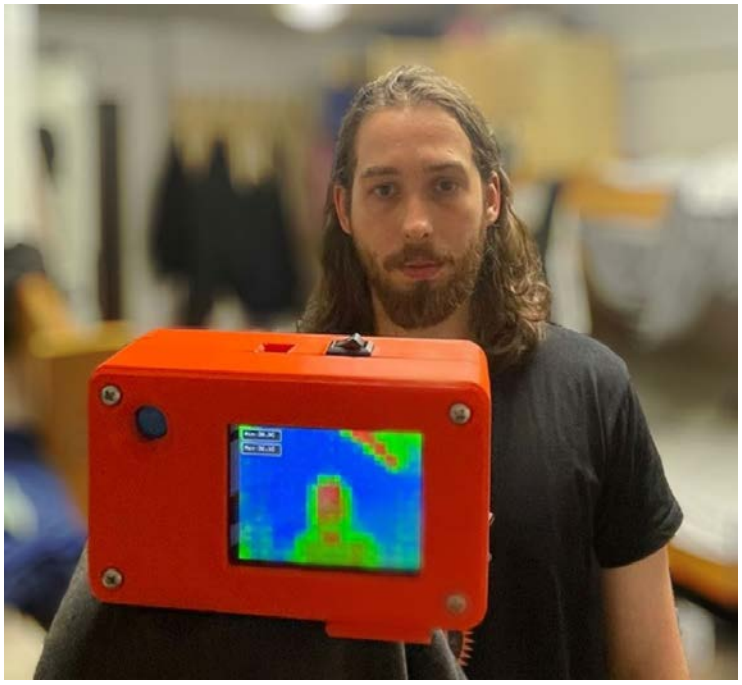


Figure 36. Testing setup for human presence in a room

This experiment can also be done in a darker room or outside with no problem because the infrared technology is not affected by the external light. It was done in a bright room for a better understanding of the project. The person is seen with a red body on the display of the camera with blue surroundings that represents the ambience which as expected is cooler than the body temperature. The temperature of the person measured by the device is 36.1°C which is a normal body temperature. In the image below (Figure 37) we can observe a picture saved from the device.

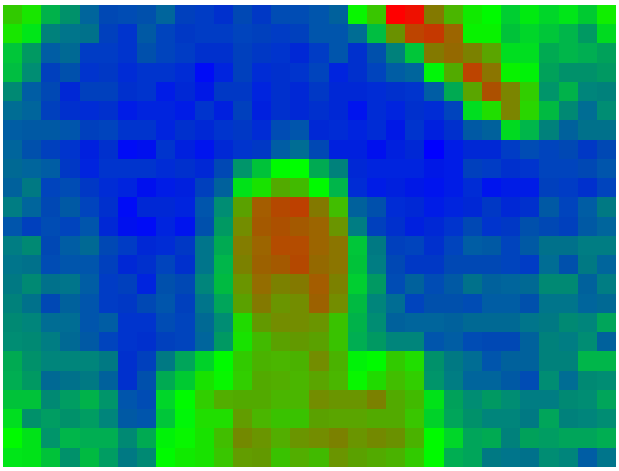


Figure 37. Thermal image taken when testing for a person presence in a room

6.4 Testing the vision in a dark medium

In order to test one of the main characteristics of the infrared camera, a test to see if the person can be detected in a dark room was done. As mentioned before, many objects, especially warm ones emit infrared radiation. This device, by using the infrared technology can still sense the presence of the person. In Figure 38 is presented how the testing was done, at night with poor lightning.



Figure 38 Testing the device in a low luminosity medium

The image recorded by the device also proves the accuracy of the sensor beacuse the person was sitting in a chair with the hands on the table separated and the picture has a separation of color at the hands separation. The red connections between the hands is a keyboard form a laptop. The saved image is in Figure 39.

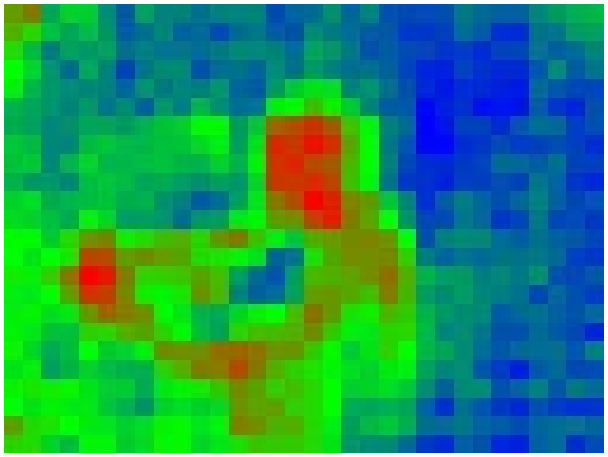


Figure 39 Result image while the subject is in a dark room

6.5 Testing Impact of Transparent Plastic Obstruction

To best evaluate any medium, a test to evaluate the impact of a barrier between the camera lens and the studied object. This test was done too on the Raspberry Pi board. As for the obstructing element, a plastic container was selected because of its optial clarity. If the barrier wouldn't be transparent like a cardboard the infrared would not be able to detect the object behind it. As a result to this experimentl, the quality of the thermal image dropped because the plastic's own emission of infrared radiation and because of scattering and refraction of the infrared radiation. These phenomenons can blur the image and affect the temperature readings. The setup simulates the scenario of having to monitor a device behind a transparent enclosure and it is presented in Figure 40.

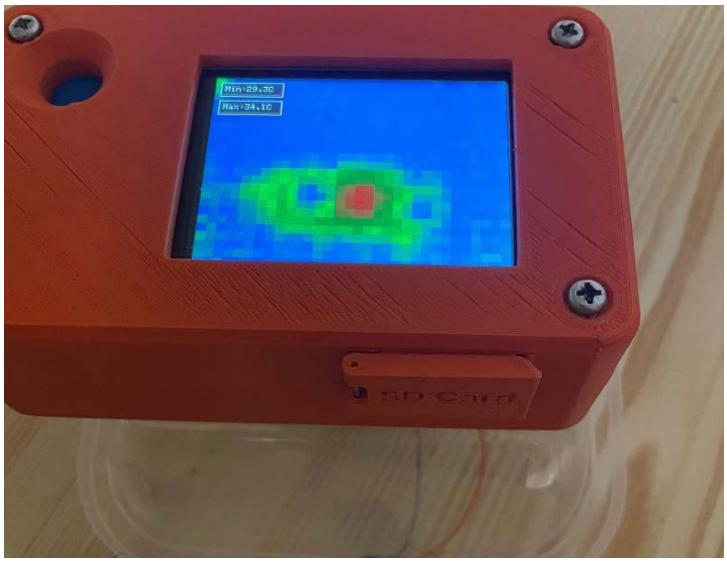


Figure 40 Testing done with a plastic obstruction in front of the subject

As for the resulting image of the experiment the image taken is in Figure 41.

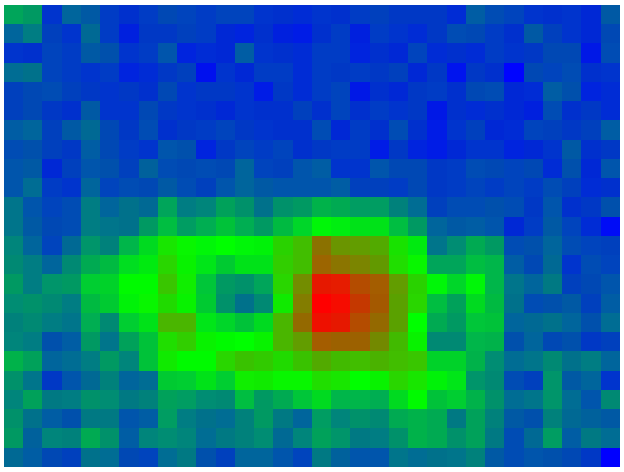


Figure 41 Result from testing with plastic obstruction

6.6 Comparing the device with an existent product

When testing the implemented system of course it had to be compared to what is already on the market to have a reference point for the results. This device was compared to the THT200 thermal imaging camera from HT instruments. Coming at more than 10 times the price of the device used of course it has more benefits. It is a high-resolution thermal camera with a resolution of 160x120 with AutoFusion and Picture in Picture functions. It also has Wi-Fi connectivity and its own application for remote usage [39]. Even with all the benefits of a already tested and manufactured device the results are quite interesting. In Figure 42 a test was done in the same way as in Figure 32.



Figure 42. Testing the Raspberry Pi zero board with the THT200 professional camera

It comes with higher accuracy and sensitivity when testing small objects but when compared with the proposed device this tool reads a maximum temperature of the main MCU of about 43.5°C and it is observed that this MCU has a nearing component that has a temperature of 35.6°C. The first device

used read a maximum of 37.6°C which is between the two values. This is happening because the first device has a much lower resolution, but the reading is not far off the actual temperature that was measured with the THT200. A solution to this problem would be to put the proposed device closer to the observed subject. When looking at smaller components the larger field of view of the used MLX90640 sensor may capture multiple parts into just one square of the checkerboard reading method. In Figure 43 the first device was moved closer to the Raspberry Pi board and a different result can be interpreted.

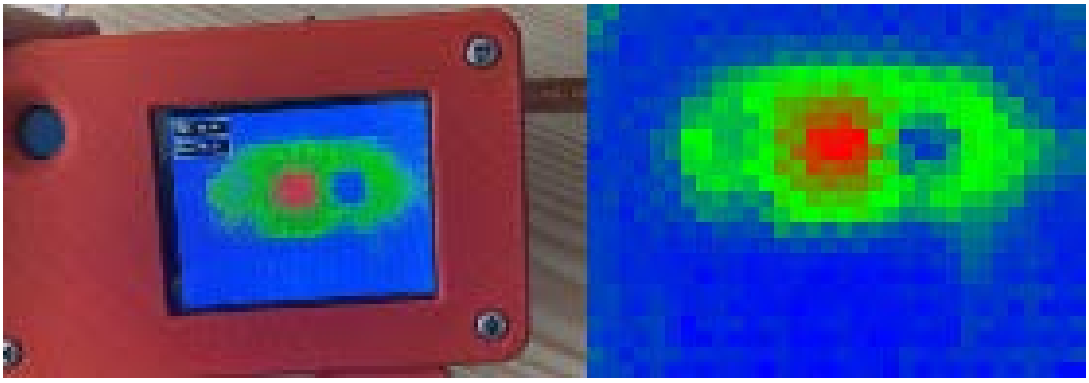


Figure 43. Result of the implemented device when reducing the range from it to the subject

When doing this test the temperature reading changes, the maximum temperature is 40.4°C which is a closer result the 43.5°C reading of the professional tool. The minimum reading here is the ambient temperature but when testing for specific components the solution for more accurate readings is to get closer to the tested subject. The precision is best in this way and gets closer and closer to the reading from the THT200 device, of course with a ±2°C accuracy error margin. Even when the device presented in this paper is in front of the object the error margin is still present the reading getting close to the actual value but with -2°C in the measurement.

Comparing the second test made in section 6.3, the THT200 gave an underwhelming result. In Figure 44 the experiment is presented.

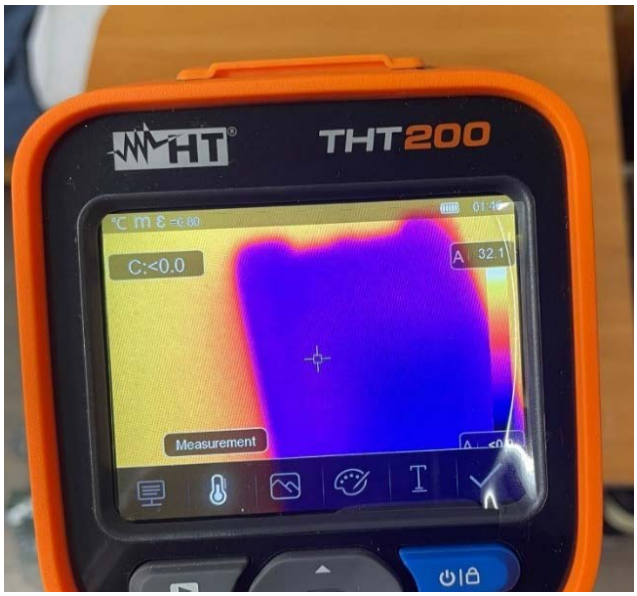


Figure 44. Testing for negative temperature reading with the THT200 tool

It is clear on the display that the ice pack is cooler than the environment, but the only result given is that its temperature is below 0°C. Here the system used gives a better result than the professional tool used.

For the third test, when detecting the presence of a human the professional tool fails again to give a better result than the first device.



Figure 45. THT200 measurement of a person

In Figure 45, the test is presented and the THT200, when pointed directly to the person, reads a temperature of 40.4°C which is specific to a fever. The person tested was in a healthy form. Again, the first device used proves to be better, especially for a variety of experiments.

The test for visibility in the dark medium was done also with the THT200, which of course did the same as the implemented device. The testing is presented in Figure 46.

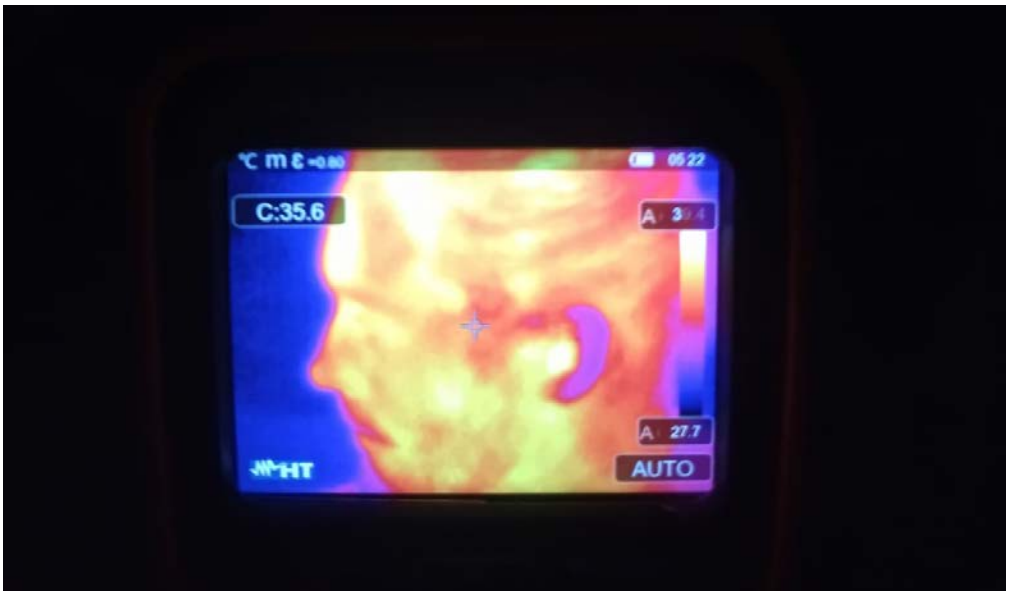


Figure 46. Test for visibility in dark environment done with THT200

When testing the performance of the THT200 camera with the transparent plastic obstruction in front of the subject, the professional tool also had problems with the thermal image quality. The reading error when testing was still at -2°C at the implemented device reading compared to the THT200, but it is harder to notice because of the distorting factors of the plastic wall between the board and the camera lens. These factors may make the reading of the MLX90640 even less accurate in some cases. Like the implemented device it displays the hot and cold objects near each other but a much lower resolution. The experiment is presented in Figure 47.

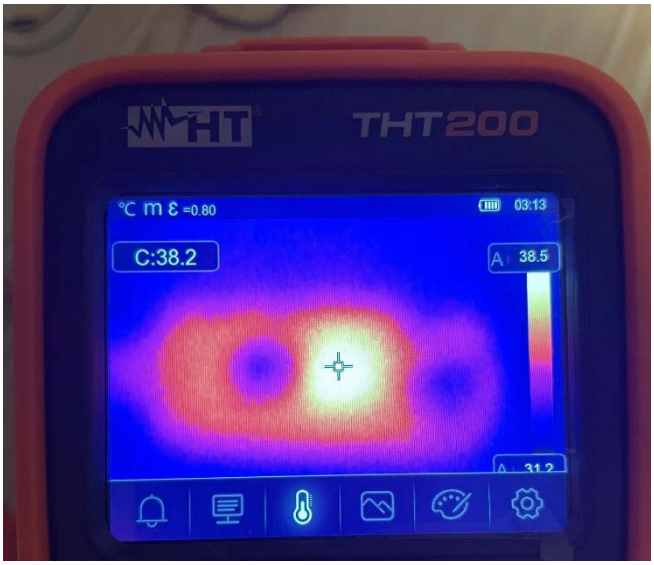


Figure 47. Result of the THT200 camera with the obstruction

While the professional tool, of course, gives a higher refresh rate of the display, a bigger resolution and a wider temperature range for measurements, when these tests are compared to the device implemented, they come as being both reliable tools. The narrow FOV of the THT200 of 20.7°x15.6° combined with the picture in picture mode make it a great tool but the results come close for the implemented device. The wider field of view with smaller resolution gives it a disadvantage when measuring small components but the test can still be done. To better understand the differences a list was made after concluding the test and analyzing the results:

- The resolution of the professional device is much higher than the one of the implemented device but they both can detect the subject by adjusting the distance from the device to the subject
- The accuracy, as expected from the datasheet of the sensor, can have a ±2°C error but in some cases the proposed device can give better results
- The precision of the discussed system is good even compared to the professional tool even if it comes at 10 times less the price
- Latency is not a problem for the device even if it has a 2Hz refresh rate because the temperature readings happen in real-time and the results prove it
- Both measuring devices come factory calibrated
- Emissivity is good while the medium is clear but when blocking the direct view of the subject both devices can make inaccurate measurements due to the infrared sensor measuring method
- The only major differences when comparing the two devices are the Picture in Picture and the AutoFusion features of the THT200 camera.

7 Conclusions

As a conclusion, the designed system is a functional thermal imaging camera with a couple of features. The successful implementation of the FireBeetle 2 ESP-32E, the MLX90640 sensor used for the thermal image formation and with the LCD TFT display the data can be observed in real time, with a high resolution for the project. The sensor used gives the opportunity to take temperature measurements for a wide range of applications as presented prior in this document. A benefit of the device is the portability of having the LiPo battery inside the casing.

The custom-designed PCB works properly and connects all the components as intended for this system. Within the layout, the SPI data lines that are separated from the rest of the LCD connections, the display works in the best possible way for this design. Also, the temperature measuring sensor communicates perfectly with the microcontroller, having no issues when using the device.

Also, the 3D printed case holds everything together, giving a professional look of the device and making it intuitive when being used by anybody. With easy access to the SD card all the thermal pictures can be later analyzed for further studies.

There is always room for improvements, especially with the power of the ESP 32 microcontroller used which gives a high opportunity for upgrades. The battery life can be extended in many ways such as software interrupts and like all the other components by upgrading the physical hardware inside the casing. Also, future improvements can be IoT integration for adding the device in a much more complex system of electrical tools and a graphical user interface for remote usage.

This approach proves that the device implemented is a suitable temperature measuring tool also when compared with professional devices and improvements can contribute in developing a high-performance system ready to be implemented in a wide range of applications.